

Chapter 3 Exercises

SOLUTIONS TO RICK DURRETT'S
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 3 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter3_Knowledge`. The cited knowledge IDs refer to entries in `chapter3_knowledge_pieces.jsonl`.

Exercise 3.1.1

Question. Generalize the proof of Lemma 3.1.1 to conclude that if

$$\max_{1 \leq j \leq n} |c_{j,n}| \rightarrow 0, \quad \sum_{j=1}^n c_{j,n} \rightarrow \lambda, \quad \sup_n \sum_{j=1}^n |c_{j,n}| < \infty,$$

then

$$\prod_{j=1}^n (1 + c_{j,n}) \rightarrow e^\lambda.$$

Knowledge used. `durrett_ch3_product_exponential_limit`;
`durrett_ch3_stirling_local_binomial`.

Solution. Let

$$M_n = \max_{1 \leq j \leq n} |c_{j,n}|, \quad A = \sup_n \sum_{j=1}^n |c_{j,n}| < \infty.$$

Since $M_n \rightarrow 0$, for all large n we have $M_n \leq 1/2$, so every $1 + c_{j,n}$ is positive and logarithms are well-defined. For $|u| \leq 1/2$,

$$|\log(1 + u) - u| \leq Cu^2$$

for an absolute constant C . Hence

$$\left| \sum_{j=1}^n \log(1 + c_{j,n}) - \sum_{j=1}^n c_{j,n} \right| \leq C \sum_{j=1}^n c_{j,n}^2 \leq CM_n \sum_{j=1}^n |c_{j,n}| \leq CAM_n \rightarrow 0.$$

Therefore

$$\sum_{j=1}^n \log(1 + c_{j,n}) \rightarrow \lambda.$$

Exponentiating gives

$$\prod_{j=1}^n (1 + c_{j,n}) = \exp \left\{ \sum_{j=1}^n \log(1 + c_{j,n}) \right\} \rightarrow e^\lambda.$$

Exercise 3.1.2

Question. The next three exercises illustrate the use of Stirling's formula. In them, $X_1, X_2, \dots$ are i.i.d. and $S_n = X_1 + \dots + X_n$.

If the X_i have a Poisson distribution with mean 1, then S_n has a Poisson distribution with mean n , i.e.,

$$\mathbb{P}(S_n = k) = e^{-n} \frac{n^k}{k!}.$$

Use Stirling's formula to show that if

$$\frac{k - n}{\sqrt{n}} \rightarrow x,$$

then

$$\sqrt{2\pi n} \mathbb{P}(S_n = k) \rightarrow \exp(-x^2/2).$$

As in the case of coin flips it follows that

$$\mathbb{P}\left(a \leq \frac{S_n - n}{\sqrt{n}} \leq b\right) \rightarrow \int_a^b (2\pi)^{-1/2} e^{-x^2/2} dx,$$

but proving the last conclusion is not part of the exercise.

Knowledge used. [durrett_ch3_stirling_local_binomial](#); [durrett_ch3_demoivre_laplace](#); [durrett_ch3_normal_approximation_continuity_correction](#).

Solution. Write

$$y_n = \frac{k - n}{n}.$$

The hypothesis says $y_n \rightarrow 0$ and $ny_n^2 \rightarrow x^2$. By Stirling's formula,

$$k! \sim \sqrt{2\pi k} k^k e^{-k}.$$

Thus

$$\begin{aligned} \sqrt{2\pi n} \mathbb{P}(S_n = k) &= \sqrt{2\pi n} e^{-n} \frac{n^k}{k!} \\ &\sim \sqrt{\frac{n}{k}} \exp\left\{-n + k + k \log \frac{n}{k}\right\}. \end{aligned}$$

Since $k = n(1 + y_n)$, the exponent is

$$-n + n(1 + y_n) + n(1 + y_n) \log \frac{1}{1 + y_n} = n\{y_n - (1 + y_n) \log(1 + y_n)\}.$$

Using

$$(1 + y) \log(1 + y) = y + \frac{y^2}{2} + O(y^3) \quad (y \rightarrow 0),$$

we obtain

$$n\{y_n - (1 + y_n) \log(1 + y_n)\} = -\frac{ny_n^2}{2} + o(1) \rightarrow -\frac{x^2}{2}.$$

Also $n/k = (1 + y_n)^{-1} \rightarrow 1$. Therefore

$$\sqrt{2\pi n} \mathbb{P}(S_n = k) \rightarrow e^{-x^2/2}.$$

Exercise 3.1.3

Question. Suppose

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}.$$

Show that if $a \in (0, 1)$, then

$$\frac{1}{2n} \log \mathbb{P}(S_{2n} \geq 2na) \rightarrow -\gamma(a),$$

where

$$\gamma(a) = \frac{1}{2} \{ (1+a) \log(1+a) + (1-a) \log(1-a) \}.$$

Knowledge used. [durrett_ch3_stirling_local_binomial](#);
[durrett_ch3_large_deviation_stirling_patterns](#).

Solution. Let

$$p_{n,j} = \mathbb{P}(S_{2n} = 2j) = \binom{2n}{n+j} 2^{-2n}, \quad -n \leq j \leq n.$$

Choose $k_n = \lceil na \rceil$. Then $\{S_{2n} \geq 2na\} = \{S_{2n} \geq 2k_n\}$, so

$$\mathbb{P}(S_{2n} \geq 2na) = \sum_{j=k_n}^n p_{n,j}.$$

First estimate the leading mass. Since $k_n/n \rightarrow a$, Stirling's formula gives

$$\begin{aligned} \frac{1}{2n} \log p_{n,k_n} &= \frac{1}{2n} \log \left\{ \binom{2n}{n+k_n} 2^{-2n} \right\} \\ &\rightarrow -\frac{1}{2} \{ (1+a) \log(1+a) + (1-a) \log(1-a) \} = -\gamma(a). \end{aligned}$$

Indeed, the logarithmic factors of order $\log n$ vanish after division by $2n$.

It remains to show that the whole tail has the same exponential rate as its first term. For $j \geq k_n$,

$$\frac{p_{n,j+1}}{p_{n,j}} = \frac{n-j}{n+j+1}.$$

Since $k_n/n \rightarrow a > 0$, for all large n and all $j \geq k_n$,

$$\frac{p_{n,j+1}}{p_{n,j}} \leq \rho \quad \text{for some fixed } \rho < 1.$$

Therefore

$$p_{n,k_n} \leq \mathbb{P}(S_{2n} \geq 2na) \leq p_{n,k_n} (1 + \rho + \rho^2 + \cdots) = \frac{p_{n,k_n}}{1 - \rho}.$$

Taking logarithms and dividing by $2n$ shows that the multiplicative constant $(1 - \rho)^{-1}$ does not affect the limit. Hence

$$\frac{1}{2n} \log \mathbb{P}(S_{2n} \geq 2na) \rightarrow -\gamma(a).$$

Exercise 3.1.4

Question. Suppose

$$\mathbb{P}(X_i = k) = e^{-1}/k!, \quad k = 0, 1, \dots$$

Show that if $a > 1$, then

$$\frac{1}{n} \log \mathbb{P}(S_n \geq na) \rightarrow a - 1 - a \log a.$$

Knowledge used. [durrett_ch3_stirling_local_binomial](#);
[durrett_ch3_large_deviation_stirling_patterns](#).

Solution. Since each X_i has the Poisson distribution with mean 1, the sum S_n has the Poisson distribution with mean n :

$$\mathbb{P}(S_n = k) = e^{-n} \frac{n^k}{k!}.$$

Let $k_n = \lceil na \rceil$. We first estimate $\mathbb{P}(S_n = k_n)$. By Stirling's formula,

$$\begin{aligned} \frac{1}{n} \log \mathbb{P}(S_n = k_n) &= \frac{1}{n} (-n + k_n \log n - \log(k_n!)) \\ &= -1 + \frac{k_n}{n} + \frac{k_n}{n} \log \frac{n}{k_n} + o(1). \end{aligned}$$

Since $k_n/n \rightarrow a$,

$$\frac{1}{n} \log \mathbb{P}(S_n = k_n) \rightarrow -1 + a - a \log a.$$

Now compare the tail with this first mass. For $j \geq k_n$,

$$\frac{\mathbb{P}(S_n = j+1)}{\mathbb{P}(S_n = j)} = \frac{n}{j+1}.$$

Because $a > 1$, for all large n and all $j \geq k_n$,

$$\frac{n}{j+1} \leq \rho \quad \text{for some fixed } \rho < 1.$$

Thus

$$\mathbb{P}(S_n = k_n) \leq \mathbb{P}(S_n \geq na) \leq \mathbb{P}(S_n = k_n)(1 + \rho + \rho^2 + \dots) = \frac{\mathbb{P}(S_n = k_n)}{1 - \rho}.$$

After taking logarithms and dividing by n , the constant factor disappears. Therefore

$$\frac{1}{n} \log \mathbb{P}(S_n \geq na) \rightarrow a - 1 - a \log a.$$

Exercise 3.2.1

Question. Give an example of random variables X_n with densities f_n so that $X_n \Rightarrow$ a uniform distribution on $(0, 1)$ but $f_n(x)$ does not converge to 1 for any $x \in [0, 1]$.

Knowledge used. [durrett_ch3_weak_convergence_definition](#);
[durrett_ch3_bounded_continuous_test_functions](#); [durrett_ch3_scheffe_theorem](#).

Solution. Let r_n be the n th Rademacher function on $[0, 1]$:

$$r_n(x) = \begin{cases} 1, & \text{the } n\text{th binary digit of } x \text{ is 1,} \\ -1, & \text{the } n\text{th binary digit of } x \text{ is 0.} \end{cases}$$

Define $f_n(x) = 1 + r_n(x)$ on points with nonterminating binary expansion. On the countable exceptional set of dyadic rationals, redefine $f_n(x) = 1 + (-1)^n/2$. This change is on a null set, so it does not change any integrals. Since $r_n = 1$ and $r_n = -1$ on sets of Lebesgue measure $1/2$, $f_n \geq 0$ and $\int_0^1 f_n(x) dx = 1$.

Let X_n have density f_n . If g is bounded and continuous, then

$$\mathbb{E}g(X_n) = \int_0^1 g(x)f_n(x) dx = \int_0^1 g(x) dx + \int_0^1 g(x)r_n(x) dx.$$

The function r_n is constant with mean 0 on each dyadic interval of length 2^{-n} . By uniform continuity of g on $[0, 1]$,

$$\int_0^1 g(x)r_n(x) dx \rightarrow 0.$$

Thus $\mathbb{E}g(X_n) \rightarrow \int_0^1 g(x) dx$, so X_n converges weakly to the uniform distribution on $(0, 1)$.

For every non-dyadic x , the binary digits are not eventually constant, so $r_n(x)$ does not converge to 0 and hence $f_n(x)$ does not converge to 1. For dyadic x , our redefinition makes $f_n(x)$ oscillate between $1 + 1/2$ and $1 - 1/2$. Therefore $f_n(x)$ does not converge to 1 for any $x \in [0, 1]$.

Exercise 3.2.2

Question. Convergence of maxima. Let $X_1, X_2, \dots$ be independent with distribution F , and let $M_n = \max_{m \leq n} X_m$. Then $\mathbb{P}(M_n \leq x) = F(x)^n$. Prove the following limit laws for M_n :

- (i) If $F(x) = 1 - x^{-\alpha}$ for $x \geq 1$, where $\alpha > 0$, then for $y > 0$,

$$\mathbb{P}(M_n/n^{1/\alpha} \leq y) \rightarrow \exp(-y^{-\alpha}).$$

- (ii) If $F(x) = 1 - |x|^\beta$ for $-1 \leq x \leq 0$, where $\beta > 0$, then for $y < 0$,

$$\mathbb{P}(n^{1/\beta} M_n \leq y) \rightarrow \exp(-|y|^\beta).$$

- (iii) If $F(x) = 1 - e^{-x}$ for $x \geq 0$, then for all $y \in \mathbb{R}$,

$$\mathbb{P}(M_n - \log n \leq y) \rightarrow \exp(-e^{-y}).$$

Knowledge used. `durrett_ch3_weak_convergence_definition`;
`durrett_ch3_continuous_mapping_theorem`; `durrett_ch3_product_exponential_limit`.

Solution. For (i), if $y > 0$, then for large n we have $n^{1/\alpha}y \geq 1$, and

$$\mathbb{P}(M_n/n^{1/\alpha} \leq y) = \left(1 - (n^{1/\alpha}y)^{-\alpha}\right)^n = \left(1 - \frac{y^{-\alpha}}{n}\right)^n \rightarrow e^{-y^{-\alpha}}.$$

For (ii), if $y < 0$, then $yn^{-1/\beta} \in [-1, 0]$ for large n . Thus

$$\mathbb{P}(n^{1/\beta} M_n \leq y) = \mathbb{P}(M_n \leq yn^{-1/\beta}) = \left(1 - \frac{|y|^\beta}{n}\right)^n \rightarrow e^{-|y|^\beta}.$$

For (iii), for large n the point $\log n + y$ is positive, and

$$\mathbb{P}(M_n - \log n \leq y) = \left(1 - e^{-(\log n + y)}\right)^n = \left(1 - \frac{e^{-y}}{n}\right)^n \rightarrow e^{-e^{-y}}.$$

Exercise 3.2.3

Question. Let $X_1, X_2, \dots$ be i.i.d. standard normal random variables.

(i) From Theorem 1.2.6, we know

$$\mathbb{P}(X_i > x) \sim \frac{1}{\sqrt{2\pi}x} e^{-x^2/2} \quad (x \rightarrow \infty).$$

Use this to conclude that for any real number θ ,

$$\frac{\mathbb{P}(X_i > x + \theta/x)}{\mathbb{P}(X_i > x)} \rightarrow e^{-\theta}.$$

(ii) Show that if b_n is defined by $\mathbb{P}(X_i > b_n) = 1/n$, then

$$\mathbb{P}(b_n(M_n - b_n) \leq x) \rightarrow \exp(-e^{-x}).$$

(iii) Show that $b_n \sim (2 \log n)^{1/2}$ and conclude that

$$\frac{M_n}{(2 \log n)^{1/2}} \rightarrow 1 \quad \text{in probability.}$$

Knowledge used. [durrett_ch3_weak_convergence_definition](#);
[durrett_ch3_product_exponential_limit](#); [durrett_ch3_converging_together_slutsky](#).

Solution. For (i), by the normal tail asymptotic,

$$\begin{aligned} \frac{\mathbb{P}(X_i > x + \theta/x)}{\mathbb{P}(X_i > x)} &\sim \frac{x}{x + \theta/x} \exp \left\{ -\frac{1}{2} \left(x + \frac{\theta}{x} \right)^2 + \frac{x^2}{2} \right\} \\ &= \frac{x}{x + \theta/x} \exp \left\{ -\theta - \frac{\theta^2}{2x^2} \right\} \rightarrow e^{-\theta}. \end{aligned}$$

For (ii),

$$\begin{aligned} \mathbb{P}(b_n(M_n - b_n) \leq x) &= \mathbb{P}(M_n \leq b_n + x/b_n) \\ &= (1 - \mathbb{P}(X_1 > b_n + x/b_n))^n. \end{aligned}$$

Since $\mathbb{P}(X_1 > b_n) = 1/n$ and (i) gives

$$\frac{\mathbb{P}(X_1 > b_n + x/b_n)}{\mathbb{P}(X_1 > b_n)} \rightarrow e^{-x},$$

we have $n\mathbb{P}(X_1 > b_n + x/b_n) \rightarrow e^{-x}$. Hence

$$(1 - \mathbb{P}(X_1 > b_n + x/b_n))^n \rightarrow \exp(-e^{-x}).$$

For (iii), the definition of b_n and the normal tail asymptotic give

$$\frac{1}{n} \sim \frac{1}{\sqrt{2\pi}b_n} e^{-b_n^2/2}.$$

Taking logarithms,

$$\log n = \frac{b_n^2}{2} + \log b_n + O(1),$$

so $b_n^2/(2 \log n) \rightarrow 1$, i.e. $b_n \sim (2 \log n)^{1/2}$.

It remains to show $M_n/b_n \rightarrow 1$ in probability. If $\varepsilon > 0$, then

$$\mathbb{P}(M_n \leq (1 - \varepsilon)b_n) = (1 - \mathbb{P}(X_1 > (1 - \varepsilon)b_n))^n \rightarrow 0,$$

because $n\mathbb{P}(X_1 > (1 - \varepsilon)b_n) \rightarrow \infty$. Also

$$\mathbb{P}(M_n > (1 + \varepsilon)b_n) \leq n\mathbb{P}(X_1 > (1 + \varepsilon)b_n) \rightarrow 0.$$

Therefore $M_n/b_n \rightarrow 1$ in probability, and since $b_n \sim (2 \log n)^{1/2}$,

$$M_n/(2 \log n)^{1/2} \rightarrow 1 \quad \text{in probability.}$$

Exercise 3.2.4

Question. Fatou's lemma. Let $g \geq 0$ be continuous. If $X_n \Rightarrow X_\infty$, then

$$\liminf_{n \rightarrow \infty} \mathbb{E}g(X_n) \geq \mathbb{E}g(X_\infty).$$

Knowledge used. [durrett_ch3_skorokhod_quantile_representation](#);
[durrett_ch3_bounded_continuous_test_functions](#).

Solution. By the almost-sure representation theorem, we may choose random variables Y_n with $Y_n \stackrel{d}{=} X_n$ and $Y_n \rightarrow Y_\infty$ almost surely, where $Y_\infty \stackrel{d}{=} X_\infty$. Since g is continuous and nonnegative, $g(Y_n) \rightarrow g(Y_\infty)$ almost surely. Fatou's lemma gives

$$\mathbb{E}g(X_\infty) = \mathbb{E}g(Y_\infty) \leq \liminf_{n \rightarrow \infty} \mathbb{E}g(Y_n) = \liminf_{n \rightarrow \infty} \mathbb{E}g(X_n).$$

Exercise 3.2.5

Question. Integration to the limit. Suppose g, h are continuous with $g(x) > 0$, and $|h(x)|/g(x) \rightarrow 0$ as $|x| \rightarrow \infty$. If $F_n \Rightarrow F$ and

$$\int g(x) dF_n(x) \leq C < \infty,$$

then

$$\int h(x) dF_n(x) \rightarrow \int h(x) dF(x).$$

Knowledge used. [durrett_ch3_bounded_continuous_test_functions](#);
[durrett_ch3_moment_tightness_criterion](#); [durrett_ch3_skorokhod_quantile_representation](#).

Solution. By Exercise 3.2.4 applied to g ,

$$\int g dF \leq \liminf_n \int g dF_n \leq C.$$

Fix $\varepsilon > 0$. Choose M so large that $|h(x)| \leq \varepsilon g(x)$ for $|x| > M$. Let ψ be a continuous cutoff with $0 \leq \psi \leq 1$, $\psi = 1$ on $[-M, M]$, and $\psi = 0$ outside $[-M - 1, M + 1]$. Then ψh is bounded and continuous, so

$$\int \psi h dF_n \rightarrow \int \psi h dF.$$

For the tails,

$$\left| \int (1 - \psi) h dF_n \right| \leq \varepsilon \int g dF_n \leq \varepsilon C,$$

and similarly

$$\left| \int (1 - \psi) h dF \right| \leq \varepsilon C.$$

Since ε is arbitrary, the desired convergence follows.

Exercise 3.2.6

Question. The Levy metric. Show that

$$\rho(F, G) = \inf \{ \varepsilon : F(x - \varepsilon) - \varepsilon \leq G(x) \leq F(x + \varepsilon) + \varepsilon \text{ for all } x \}$$

defines a metric on the space of distributions and $\rho(F_n, F) \rightarrow 0$ if and only if $F_n \Rightarrow F$.

Knowledge used. [durrett_ch3_weak_convergence_definition](#);
[durrett_ch3_portmanteau_open_closed](#); [durrett_ch3_helly_selection_tightness](#).

Solution. The quantity $\rho(F, G)$ is nonnegative. The defining inequalities are symmetric in F and G after using monotonicity and right-continuity of distribution functions, so $\rho(F, G) = \rho(G, F)$. If $\rho(F, G) = 0$, then for every $\varepsilon > 0$,

$$F(x - \varepsilon) - \varepsilon \leq G(x) \leq F(x + \varepsilon) + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ at continuity points of F gives $G(x) = F(x)$ there, and right-continuity gives $F = G$. Conversely $F = G$ clearly implies $\rho(F, G) = 0$.

For the triangle inequality, suppose $\rho(F, G) < \varepsilon$ and $\rho(G, H) < \delta$. Then

$$\begin{aligned} H(x) &\leq G(x + \delta) + \delta \leq F(x + \delta + \varepsilon) + \varepsilon + \delta, \\ H(x) &\geq G(x - \delta) - \delta \geq F(x - \delta - \varepsilon) - \varepsilon - \delta. \end{aligned}$$

Thus $\rho(F, H) \leq \varepsilon + \delta$, and taking infima gives the triangle inequality.

If $\rho(F_n, F) \rightarrow 0$ and x is a continuity point of F , then for every $\varepsilon > 0$ and all large n ,

$$F(x - \varepsilon) - \varepsilon \leq F_n(x) \leq F(x + \varepsilon) + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives $F_n(x) \rightarrow F(x)$, so $F_n \Rightarrow F$.

Conversely, suppose $F_n \Rightarrow F$. Fix $\varepsilon > 0$. Choose continuity points

$$-\infty = x_0 < x_1 < \cdots < x_m = \infty$$

with $F(x_i) - F(x_{i-1}) < \varepsilon$ for the finite intervals and with the two tails smaller than ε . Since $F_n(x_i) \rightarrow F(x_i)$ at the finite grid points, for all large n the Levy inequalities hold with a constant multiple of ε . As ε is arbitrary, $\rho(F_n, F) \rightarrow 0$.

Exercise 3.2.7

Question. The Ky Fan metric on random variables is defined by

$$\alpha(X, Y) = \inf\{\varepsilon \geq 0 : \mathbb{P}(|X - Y| > \varepsilon) \leq \varepsilon\}.$$

Show that if $\alpha(X, Y) = \alpha$, then the corresponding distributions have Levy distance $\rho(F, G) \leq \alpha$.

Knowledge used. [durrett_ch3_weak_convergence_definition](#);
[durrett_ch3_convergence_in_probability_to_distribution](#).

Solution. Let $\varepsilon > \alpha$. Then

$$\mathbb{P}(|X - Y| > \varepsilon) \leq \varepsilon.$$

For every x ,

$$G(x) = \mathbb{P}(Y \leq x) \leq \mathbb{P}(X \leq x + \varepsilon) + \mathbb{P}(|X - Y| > \varepsilon) \leq F(x + \varepsilon) + \varepsilon,$$

$$G(x) = \mathbb{P}(Y \leq x) \geq \mathbb{P}(X \leq x - \varepsilon) - \mathbb{P}(|X - Y| > \varepsilon) \geq F(x - \varepsilon) - \varepsilon.$$

Thus $\rho(F, G) \leq \varepsilon$. Letting $\varepsilon \downarrow \alpha$ gives $\rho(F, G) \leq \alpha$.

Exercise 3.2.8

Question. Let $\alpha(X, Y)$ be the metric in the previous exercise and let

$$\beta(X, Y) = \mathbb{E} \left(\frac{|X - Y|}{1 + |X - Y|} \right)$$

be the metric of Exercise 2.3.6. If $\alpha(X, Y) = a$, then

$$\frac{a^2}{1 + a} \leq \beta(X, Y) \leq a + \frac{(1 - a)a}{1 + a}.$$

Knowledge used. [durrett_ch3_convergence_in_probability_to_distribution](#);
[durrett_ch3_weak_convergence_definition](#).

Solution. Put $Z = |X - Y|$. Since $\alpha(X, Y) \leq 1$, we have $0 \leq a \leq 1$. For every $\varepsilon > a$,

$$\mathbb{P}(Z > \varepsilon) \leq \varepsilon.$$

Therefore

$$\begin{aligned} \beta(X, Y) &= \mathbb{E} \frac{Z}{1 + Z} \\ &\leq \mathbb{P}(Z > \varepsilon) + \frac{\varepsilon}{1 + \varepsilon} \mathbb{P}(Z \leq \varepsilon) \\ &\leq \varepsilon + \frac{\varepsilon}{1 + \varepsilon} (1 - \varepsilon). \end{aligned}$$

Letting $\varepsilon \downarrow a$ gives the upper bound.

For the lower bound, if $0 < b < a$, then by the definition of the infimum,

$$\mathbb{P}(Z > b) > b.$$

Hence

$$\beta(X, Y) \geq \mathbb{E} \left(\frac{Z}{1 + Z}; Z > b \right) \geq \frac{b}{1 + b} \mathbb{P}(Z > b) > \frac{b^2}{1 + b}.$$

Letting $b \uparrow a$ gives

$$\beta(X, Y) \geq \frac{a^2}{1 + a}.$$

Exercise 3.2.9

Question. If $F_n \Rightarrow F$ and F is continuous, then

$$\sup_x |F_n(x) - F(x)| \rightarrow 0.$$

Knowledge used. [durrett_ch3_weak_convergence_definition](#);
[durrett_ch3_portmanteau_open_closed](#).

Solution. Fix $\varepsilon > 0$. Since F is continuous and has limits 0 and 1 at $\pm\infty$, choose points

$$-\infty = x_0 < x_1 < \cdots < x_m = \infty$$

so that $F(x_i) - F(x_{i-1}) < \varepsilon$ for each finite interval and the two tails are smaller than ε . Since every finite x_i is a continuity point of F , $F_n(x_i) \rightarrow F(x_i)$.

For all large n , $|F_n(x_i) - F(x_i)| < \varepsilon$ at all finite grid points. If $x_{i-1} \leq x \leq x_i$, monotonicity gives

$$F_n(x_{i-1}) \leq F_n(x) \leq F_n(x_i), \quad F(x_{i-1}) \leq F(x) \leq F(x_i).$$

Using the grid bounds, $|F_n(x) - F(x)| \leq 3\varepsilon$ uniformly in x . Since ε is arbitrary, the supremum tends to 0.

Exercise 3.2.10

Question. If F is any distribution function, there is a sequence of distribution functions of the form

$$\sum_{m=1}^n a_{n,m} \mathbf{1}(x_{n,m} \leq x)$$

with $F_n \Rightarrow F$.

Knowledge used. [durrett_ch3_skorokhod_quantile_representation](#);
[durrett_ch3_weak_convergence_definition](#).

Solution. Let U be uniform on $(0, 1)$ and let

$$Q(u) = \sup\{x : F(x) < u\}$$

be the quantile function. Then $Q(U)$ has distribution F . Define

$$U_n = \frac{\lceil nU \rceil - 1/2}{n}, \quad X_n = Q(U_n).$$

Then X_n takes only the finitely many values $Q(1/(2n)), Q(3/(2n)), \dots, Q((2n-1)/(2n))$, so its distribution function has the required form, with weights $a_{n,m} = 1/n$ after combining equal atoms if necessary.

For all u at which Q is continuous, $U_n \rightarrow u$ implies $Q(U_n) \rightarrow Q(u)$. The discontinuities of a monotone function are countable, so $X_n \rightarrow Q(U)$ almost surely. Hence $X_n \Rightarrow Q(U)$, whose distribution is F .

Exercise 3.2.11

Question. Let X_n , $1 \leq n \leq \infty$, be integer valued. Show that $X_n \Rightarrow X_\infty$ if and only if

$$\mathbb{P}(X_n = m) \rightarrow \mathbb{P}(X_\infty = m) \quad \text{for all } m \in \mathbb{Z}.$$

Knowledge used. [durrett_ch3_weak_convergence_definition](#);
[durrett_ch3_portmanteau_open_closed](#); [durrett_ch3_scheffe_theorem](#).

Solution. Suppose first that $X_n \Rightarrow X_\infty$. For an integer m ,

$$\mathbb{P}(X_n = m) = \mathbb{P}(X_n \leq m + 1/2) - \mathbb{P}(X_n \leq m - 1/2).$$

The points $m \pm 1/2$ are continuity points of the distribution function of X_∞ , so the right side converges to $\mathbb{P}(X_\infty = m)$.

Conversely, write

$$p_n(m) = \mathbb{P}(X_n = m), \quad p(m) = \mathbb{P}(X_\infty = m).$$

Pointwise convergence $p_n(m) \rightarrow p(m)$ and $\sum_m p_n(m) = \sum_m p(m) = 1$ imply

$$\sum_{m \in \mathbb{Z}} |p_n(m) - p(m)| \rightarrow 0$$

by Scheffe's theorem for counting measure. Hence $\mathbb{P}(X_n \in A) \rightarrow \mathbb{P}(X_\infty \in A)$ for every $A \subseteq \mathbb{Z}$, in particular for half-lines. Therefore $X_n \Rightarrow X_\infty$.

Exercise 3.2.12

Question. Show that if $X_n \rightarrow X$ in probability, then $X_n \Rightarrow X$ and that, conversely, if $X_n \Rightarrow c$, where c is a constant, then $X_n \rightarrow c$ in probability.

Knowledge used. [durrett_ch3_convergence_in_probability_to_distribution](#);
[durrett_ch3_portmanteau_open_closed](#).

Solution. Assume $X_n \rightarrow X$ in probability. Let x be a continuity point of the distribution function F of X . For $\varepsilon > 0$,

$$\mathbb{P}(X_n \leq x) \leq \mathbb{P}(X \leq x + \varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon),$$

and

$$\mathbb{P}(X_n \leq x) \geq \mathbb{P}(X \leq x - \varepsilon) - \mathbb{P}(|X_n - X| > \varepsilon).$$

Let $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ to get $\mathbb{P}(X_n \leq x) \rightarrow F(x)$. Hence $X_n \Rightarrow X$.

Conversely, suppose $X_n \Rightarrow c$. If $\varepsilon > 0$, then $c - \varepsilon$ and $c + \varepsilon$ are continuity points of the distribution function of the constant c . Thus

$$\mathbb{P}(|X_n - c| > \varepsilon) \leq \mathbb{P}(X_n \leq c - \varepsilon) + \mathbb{P}(X_n > c + \varepsilon) \rightarrow 0.$$

Therefore $X_n \rightarrow c$ in probability.

Exercise 3.2.13

Question. Converging together lemma. If $X_n \Rightarrow X$ and $Y_n \Rightarrow c$, where c is a constant, then

$$X_n + Y_n \Rightarrow X + c.$$

A useful consequence is that if $X_n \Rightarrow X$ and $Z_n - X_n \Rightarrow 0$, then $Z_n \Rightarrow X$.

Knowledge used. [durrett_ch3_converging_together_slutsky](#);
[durrett_ch3_convergence_in_probability_to_distribution](#);
[durrett_ch3_continuous_mapping_theorem](#).

Solution. By Exercise 3.2.12, $Y_n \rightarrow c$ in probability. Let f be bounded and continuous. Since $X_n \Rightarrow X$, the sequence X_n is tight. Pick M so that $\mathbb{P}(|X_n| > M)$ is small uniformly for large n . On $[-M-1, M+1]$, f is uniformly continuous, so

$$f(X_n + Y_n) - f(X_n + c) \rightarrow 0 \quad \text{in probability,}$$

and boundedness makes the expectations converge to 0. Also $X_n + c \Rightarrow X + c$ by the continuous mapping theorem. Hence $X_n + Y_n \Rightarrow X + c$.

For the consequence, put $Y_n = Z_n - X_n$. Then $Y_n \Rightarrow 0$, so

$$Z_n = X_n + (Z_n - X_n) \Rightarrow X.$$

Exercise 3.2.14

Question. Suppose $X_n \Rightarrow X$, $Y_n \geq 0$, and $Y_n \Rightarrow c$, where $c > 0$ is a constant. Show that

$$X_n Y_n \Rightarrow cX.$$

Knowledge used. [durrett_ch3_converging_together_slutsky](#);
[durrett_ch3_convergence_in_probability_to_distribution](#);
[durrett_ch3_continuous_mapping_theorem](#).

Solution. Since $Y_n \Rightarrow c$ and c is constant, Exercise 3.2.12 gives $Y_n \rightarrow c$ in probability. The sequence X_n is tight because it converges in distribution. Thus for every $\varepsilon > 0$, choose M large so that $\mathbb{P}(|X_n| > M)$ is small for large n . Then

$$\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \leq \mathbb{P}(|X_n| > M) + \mathbb{P}(|Y_n - c| > \varepsilon/M) \rightarrow 0.$$

So $X_n(Y_n - c) \rightarrow 0$ in probability, hence in distribution. Since $cX_n \Rightarrow cX$ by the continuous mapping theorem, the converging together lemma gives

$$X_n Y_n = cX_n + X_n(Y_n - c) \Rightarrow cX.$$

Exercise 3.2.15

Question. Show that if

$$X_n = (X_n^1, \dots, X_n^n)$$

is uniformly distributed over the surface of the sphere of radius $\sqrt{n}$ in $\mathbb{R}^n$, then $X_n^1 \Rightarrow$ a standard normal. Hint: Let $Y_1, Y_2, \dots$ be i.i.d. standard normals and let

$$X_n^i = Y_i \left(\frac{n}{\sum_{m=1}^n Y_m^2} \right)^{1/2}.$$

Knowledge used. [durrett_ch3_converging_together_slutsky](#);
[durrett_ch3_continuous_mapping_theorem](#); [durrett_ch3_iid_clt](#).

Solution. Let $Y_1, Y_2, \dots$ be i.i.d. standard normals. By rotational symmetry of the multivariate normal vector,

$$\sqrt{n} \frac{(Y_1, \dots, Y_n)}{(\sum_{m=1}^n Y_m^2)^{1/2}}$$

is uniformly distributed on the sphere of radius $\sqrt{n}$. Therefore we may represent

$$X_n^1 = Y_1 \left(\frac{n}{\sum_{m=1}^n Y_m^2} \right)^{1/2} = \frac{Y_1}{(n^{-1} \sum_{m=1}^n Y_m^2)^{1/2}}.$$

By the law of large numbers,

$$n^{-1} \sum_{m=1}^n Y_m^2 \rightarrow \mathbb{E}Y_1^2 = 1 \quad \text{in probability.}$$

Hence the denominator converges to 1 in probability. Since Y_1 has the standard normal distribution, Slutsky's theorem gives

$$X_n^1 \Rightarrow Y_1,$$

the standard normal law.

Exercise 3.2.16

Question. Suppose $Y_n \geq 0$,

$$\mathbb{E}Y_n^\alpha \rightarrow 1 \quad \text{and} \quad \mathbb{E}Y_n^\beta \rightarrow 1$$

for some $0 < \alpha < \beta$. Show that $Y_n \rightarrow 1$ in probability.

Knowledge used. [durrett_ch3_convergence_in_probability_to_distribution](#);
[durrett_ch3_moment_tightness_criterion](#).

Solution. Put $Z_n = Y_n^\alpha$ and $p = \beta/\alpha > 1$. Then

$$\mathbb{E}Z_n \rightarrow 1, \quad \mathbb{E}Z_n^p \rightarrow 1.$$

The convex function

$$q(z) = z^p - 1 - p(z - 1), \quad z \geq 0,$$

is nonnegative and equals 0 only at $z = 1$. Moreover,

$$\mathbb{E}q(Z_n) = \mathbb{E}Z_n^p - 1 - p(\mathbb{E}Z_n - 1) \rightarrow 0.$$

For any $\delta > 0$,

$$\eta_\delta = \inf_{|z-1| \geq \delta, z \geq 0} q(z) > 0,$$

so

$$\mathbb{P}(|Z_n - 1| \geq \delta) \leq \frac{\mathbb{E}q(Z_n)}{\eta_\delta} \rightarrow 0.$$

Thus $Z_n \rightarrow 1$ in probability. Since $y \mapsto y^{1/\alpha}$ is continuous on $[0, \infty)$, $Y_n = Z_n^{1/\alpha} \rightarrow 1$ in probability.

Exercise 3.2.17

Question. For each $K < \infty$ and $y < 1$ there is a $c_{y,K} > 0$ so that

$$\mathbb{E}X^2 = 1 \quad \text{and} \quad \mathbb{E}X^4 \leq K$$

implies

$$\mathbb{P}(|X| > y) \geq c_{y,K}.$$

Knowledge used. [durrett_ch3_moment_tightness_criterion](#);
[durrett_ch3_portmanteau_open_closed](#).

Solution. Let $A = \{|X| > y\}$, $p = \mathbb{P}(A)$, and $y_+ = \max(y, 0)$. Since $\mathbb{E}X^2 = 1$,

$$1 = \mathbb{E}(X^2; A^c) + \mathbb{E}(X^2; A) \leq y_+^2 + \mathbb{E}(X^2; A).$$

By Cauchy's inequality,

$$\mathbb{E}(X^2; A) \leq (\mathbb{E}X^4)^{1/2} \mathbb{P}(A)^{1/2} \leq K^{1/2} p^{1/2}.$$

Therefore

$$1 \leq y_+^2 + K^{1/2} p^{1/2},$$

and hence

$$p \geq \frac{(1 - y_+^2)^2}{K}.$$

Thus one may take

$$c_{y,K} = \frac{(1 - y_+^2)^2}{K} > 0.$$

Exercise 3.3.1

Question. Show that if ϕ is a characteristic function, then $\operatorname{Re} \phi$ and $|\phi|^2$ are also characteristic functions.

Knowledge used. [durrett_ch3_characteristic_function_definition](#);
[durrett_ch3_cf_independent_sums](#).

Solution. Let X have characteristic function ϕ . Then $-X$ has characteristic function $\phi(-t) = \overline{\phi(t)}$. If ϵ is independent of X and takes values ± 1 with probability $1/2$, then ϵX has characteristic function

$$\frac{\phi(t) + \phi(-t)}{2} = \operatorname{Re} \phi(t).$$

Thus $\operatorname{Re} \phi$ is a characteristic function.

Let X' be an independent copy of X . The characteristic function of $X - X'$ is

$$\phi(t)\phi(-t) = |\phi(t)|^2.$$

Hence $|\phi|^2$ is also a characteristic function.

Exercise 3.3.2

Question.

(i) Show that

$$\mu(\{a\}) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-ita} \phi(t) dt.$$

(ii) If $\mathbb{P}(X \in h\mathbb{Z}) = 1$, $h > 0$, then $\phi(2\pi/h + t) = \phi(t)$ and

$$\mathbb{P}(X = x) = \frac{h}{2\pi} \int_{-\pi/h}^{\pi/h} e^{-itx} \phi(t) dt, \quad x \in h\mathbb{Z}.$$

(iii) If $X = Y + b$, then $\mathbb{E}e^{itX} = e^{itb}\mathbb{E}e^{itY}$. Hence if $\mathbb{P}(X \in b + h\mathbb{Z}) = 1$, the formula in (ii) is valid for $x \in b + h\mathbb{Z}$.

Knowledge used. [durrett_ch3_cf_inversion_formula](#);
[durrett_ch3_characteristic_function_definition](#).

Solution. For (i), by Fubini,

$$\frac{1}{2T} \int_{-T}^T e^{-ita} \phi(t) dt = \int \frac{1}{2T} \int_{-T}^T e^{it(x-a)} dt \mu(dx).$$

The inner average equals 1 if $x = a$ and

$$\frac{\sin(T(x-a))}{T(x-a)}$$

if $x \neq a$. It is bounded in absolute value by 1 and converges pointwise to $\mathbf{1}_{\{a\}}(x)$. Dominated convergence gives the formula.

For (ii), if $X \in h\mathbb{Z}$, then $e^{i(2\pi/h)X} = 1$ almost surely, so $\phi(t + 2\pi/h) = \phi(t)$. For $x \in h\mathbb{Z}$,

$$\frac{h}{2\pi} \int_{-\pi/h}^{\pi/h} e^{-itx} \phi(t) dt = \sum_{y \in h\mathbb{Z}} \mathbb{P}(X = y) \frac{h}{2\pi} \int_{-\pi/h}^{\pi/h} e^{it(y-x)} dt.$$

The last integral is 1 when $y = x$ and 0 otherwise, so the expression equals $\mathbb{P}(X = x)$.

Part (iii) follows immediately from

$$\mathbb{E}e^{itX} = \mathbb{E}e^{it(Y+b)} = e^{itb}\mathbb{E}e^{itY}.$$

Apply (ii) to $Y = X - b$, which is supported on $h\mathbb{Z}$.

Exercise 3.3.3

Question. Suppose X and Y are independent and have characteristic function ϕ and distribution μ . Apply Exercise 3.3.2 to $X - Y$ and use Exercise 2.1.5 to get

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |\phi(t)|^2 dt = \mathbb{P}(X - Y = 0) = \sum_x \mu(\{x\})^2.$$

Knowledge used. [durrett_ch3_cf_independent_sums](#); [durrett_ch3_cf_inversion_formula](#); [durrett_ch2_joint_law_product_measure](#).

Solution. The characteristic function of $X - Y$ is

$$\mathbb{E}e^{it(X-Y)} = \phi(t)\phi(-t) = |\phi(t)|^2.$$

Exercise 3.3.2(i), with $a = 0$, gives

$$\mathbb{P}(X - Y = 0) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |\phi(t)|^2 dt.$$

Since X and Y are independent with common law μ ,

$$\mathbb{P}(X - Y = 0) = \sum_x \mathbb{P}(X = x, Y = x) = \sum_x \mu(\{x\})^2,$$

where the sum is over the at most countable set of atoms of μ .

Exercise 3.3.4

Question. Give an example of a measure μ with a density but for which

$$\int |\phi(t)| dt = \infty.$$

Knowledge used. [durrett_ch3_standard_characteristic_functions](#); [durrett_ch3_integrable_cf_density](#).

Solution. Let μ be the uniform distribution on $(-1, 1)$. Its density is $f(x) = 1/2$ on $(-1, 1)$ and its characteristic function is

$$\phi(t) = \frac{\sin t}{t}.$$

Then

$$\int_{\mathbb{R}} |\phi(t)| dt = 2 \int_0^\infty \frac{|\sin t|}{t} dt = \infty,$$

because the integral of $|\sin t|/t$ diverges like a constant multiple of the harmonic series over intervals $[k\pi, (k+1)\pi]$.

Exercise 3.3.5

Question. Show that if $X_1, \dots, X_n$ are independent and uniformly distributed on $(-1, 1)$, then for $n \geq 2$, $X_1 + \dots + X_n$ has density

$$f(x) = \frac{1}{\pi} \int_0^\infty \left(\frac{\sin t}{t} \right)^n \cos(tx) dt.$$

Knowledge used. [durrett_ch3_standard_characteristic_functions](#); [durrett_ch3_cf_independent_sums](#); [durrett_ch3_integrable_cf_density](#).

Solution. For X uniform on $(-1, 1)$,

$$\phi_X(t) = \frac{1}{2} \int_{-1}^1 e^{itx} dx = \frac{\sin t}{t}.$$

By independence, $S_n = X_1 + \cdots + X_n$ has characteristic function

$$\phi_{S_n}(t) = \left(\frac{\sin t}{t} \right)^n.$$

For $n \geq 2$, this function is integrable, since it is bounded near 0 and is $O(t^{-n})$ at infinity. Fourier inversion for integrable characteristic functions gives

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \left(\frac{\sin t}{t} \right)^n dt.$$

The imaginary part is odd and the real part is even, so this becomes

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \left(\frac{\sin t}{t} \right)^n \cos(tx) dt.$$

Exercise 3.3.6

Question. Use the result in Example 3.3.16 to conclude that if $X_1, X_2, \dots$ are independent and have the Cauchy distribution, then

$$\frac{X_1 + \cdots + X_n}{n}$$

has the same distribution as X_1 .

Knowledge used. [durrett_ch3_standard_characteristic_functions](#);
[durrett_ch3_cf_independent_sums](#); [durrett_ch3_levy_continuity_theorem](#).

Solution. The standard Cauchy characteristic function is

$$\phi(t) = e^{-|t|}.$$

If $S_n = X_1 + \cdots + X_n$, then

$$\phi_{S_n/n}(t) = \prod_{j=1}^n \phi(t/n) = \left(e^{-|t|/n} \right)^n = e^{-|t|}.$$

Thus S_n/n and X_1 have the same characteristic function, hence the same distribution.

Exercise 3.3.7

Question. Suppose that $X_n \Rightarrow X$ and X_n has a normal distribution with mean 0 and variance σ_n^2 . Prove that

$$\sigma_n^2 \rightarrow \sigma^2 \in [0, \infty).$$

Knowledge used. [durrett_ch3_standard_characteristic_functions](#);
[durrett_ch3_levy_continuity_theorem](#); [durrett_ch3_subsequence_principle_weak](#).

Solution. The characteristic function of X_n is

$$\phi_n(t) = \exp(-\sigma_n^2 t^2 / 2).$$

Since $X_n \Rightarrow X$, $\phi_n(t) \rightarrow \phi(t)$ for every t . At $t = 1$,

$$e^{-\sigma_n^2 / 2} \rightarrow \phi(1).$$

The limit cannot be 0. If it were, then $\sigma_n^2 \rightarrow \infty$ along a subsequence, and the normal laws would not be tight, contradicting weak convergence. Hence $\phi(1) > 0$, and

$$\sigma_n^2 \rightarrow -2 \log \phi(1) \equiv \sigma^2 < \infty.$$

Clearly $\sigma^2 \geq 0$.

Exercise 3.3.8

Question. Show that if X_n and Y_n are independent for $1 \leq n \leq \infty$, $X_n \Rightarrow X_\infty$, and $Y_n \Rightarrow Y_\infty$, then

$$X_n + Y_n \Rightarrow X_\infty + Y_\infty.$$

Knowledge used. [durrett_ch3_cf_independent_sums](#); [durrett_ch3_levy_continuity_theorem](#).

Solution. Let ϕ_n, ψ_n be the characteristic functions of X_n, Y_n . By the continuity theorem,

$$\phi_n(t) \rightarrow \phi_\infty(t), \quad \psi_n(t) \rightarrow \psi_\infty(t).$$

Since X_n and Y_n are independent,

$$\phi_{X_n+Y_n}(t) = \phi_n(t)\psi_n(t) \rightarrow \phi_\infty(t)\psi_\infty(t).$$

The right-hand side is the characteristic function of $X_\infty + Y_\infty$, because X_∞ and Y_∞ are independent. The Levy continuity theorem gives the desired weak convergence.

Exercise 3.3.9

Question. Let $X_1, X_2, \dots$ be independent and let $S_n = X_1 + \dots + X_n$. Let ϕ_j be the characteristic function of X_j and suppose that $S_n \rightarrow S_\infty$ almost surely. Then S_∞ has characteristic function

$$\prod_{j=1}^{\infty} \phi_j(t).$$

Knowledge used. [durrett_ch3_cf_independent_sums](#); [durrett_ch3_convergence_in_probability_to_distribution](#); [durrett_ch3_levy_continuity_theorem](#).

Solution. For each fixed n , independence gives

$$\phi_{S_n}(t) = \prod_{j=1}^n \phi_j(t).$$

Since $S_n \rightarrow S_\infty$ almost surely, $S_n \Rightarrow S_\infty$. Therefore by the continuity theorem,

$$\prod_{j=1}^n \phi_j(t) = \phi_{S_n}(t) \rightarrow \phi_{S_\infty}(t).$$

Thus the infinite product converges pointwise and equals the characteristic function of S_∞ .

Exercise 3.3.10

Question. Using the identity

$$\sin t = 2 \sin(t/2) \cos(t/2)$$

repeatedly leads to

$$\frac{\sin t}{t} = \prod_{m=1}^{\infty} \cos(t/2^m).$$

Prove the last identity by interpreting each side as a characteristic function.

Knowledge used. [durrett_ch3_standard_characteristic_functions](#);
[durrett_ch3_cf_independent_sums](#); [durrett_ch3_levy_continuity_theorem](#).

Solution. Let $\epsilon_1, \epsilon_2, \dots$ be independent random variables with $\mathbb{P}(\epsilon_m = 1) = \mathbb{P}(\epsilon_m = -1) = 1/2$. Then $2^{-m}\epsilon_m$ has characteristic function $\cos(t/2^m)$. The series

$$Y = \sum_{m=1}^{\infty} 2^{-m}\epsilon_m$$

converges almost surely. By Exercise 3.3.9, its characteristic function is

$$\prod_{m=1}^{\infty} \cos(t/2^m).$$

But Y is uniformly distributed on $[-1, 1]$: its binary expansion gives the usual dyadic construction of the uniform law. Hence its characteristic function is

$$\frac{1}{2} \int_{-1}^1 e^{itx} dx = \frac{\sin t}{t}.$$

Therefore

$$\frac{\sin t}{t} = \prod_{m=1}^{\infty} \cos(t/2^m).$$

Exercise 3.3.11

Question. Let $X_1, X_2, \dots$ be independent, taking values 0 and 1 with probability $1/2$ each. The random variable

$$X = 2 \sum_{j \geq 1} \frac{X_j}{3^j}$$

has the Cantor distribution. Compute the characteristic function ϕ of X and notice that ϕ has the same value at $t = 3^k \pi$ for $k = 0, 1, 2, \dots$

Knowledge used. [durrett_ch3_cf_independent_sums](#);
[durrett_ch3_characteristic_function_definition](#).

Solution. The characteristic function of $2X_j/3^j$ is

$$\frac{1}{2} (1 + e^{i2t/3^j}) = e^{it/3^j} \cos(t/3^j).$$

By independence and almost sure convergence of the series,

$$\phi(t) = \prod_{j=1}^{\infty} \frac{1}{2} (1 + e^{i2t/3^j}) = e^{it/2} \prod_{j=1}^{\infty} \cos(t/3^j),$$

because $\sum_{j \geq 1} 3^{-j} = 1/2$.

If $t = 3^k \pi$, then one factor is

$$\cos(3^k \pi / 3^k) = \cos \pi = -1$$

and for $j < k$ the factors are $\cos(3^{k-j} \pi) = -1$, while for $j > k$ the remaining tail is the same as the tail at π . More simply, the product formula gives

$$\phi(3^k \pi) = e^{i3^k \pi / 2} \prod_{j=1}^{\infty} \cos(3^k \pi / 3^j) = e^{i\pi / 2} \prod_{j=1}^{\infty} \cos(\pi / 3^j) = \phi(\pi),$$

since the extra finite factors and exponential phase cancel. Thus $\phi(3^k \pi)$ has the same value for all $k \geq 0$.

Exercise 3.3.12

Question. Use Theorem 3.3.18 and the series expansion for $e^{-t^2/2}$ to show that the standard normal distribution has

$$\mathbb{E}X^{2n} = \frac{(2n)!}{2^n n!} = (2n-1)(2n-3) \cdots 3 \cdot 1 \equiv (2n-1)!!.$$

Knowledge used. [durrett_ch3_derivatives_moments](#);
[durrett_ch3_standard_characteristic_functions](#).

Solution. For a standard normal X ,

$$\phi(t) = \mathbb{E}e^{itX} = e^{-t^2/2} = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{2^k k!}.$$

On the other hand, using moments as derivatives,

$$\phi(t) = \sum_{m=0}^{\infty} \frac{i^m \mathbb{E}X^m}{m!} t^m$$

as a formal Taylor expansion at 0. Comparing the coefficient of t^{2n} gives

$$\frac{i^{2n} \mathbb{E}X^{2n}}{(2n)!} = \frac{(-1)^n}{2^n n!}.$$

Since $i^{2n} = (-1)^n$,

$$\mathbb{E}X^{2n} = \frac{(2n)!}{2^n n!}.$$

The last expression equals $(2n-1)!!$ by canceling the even factors in $(2n)!$.

Exercise 3.3.13

Question.

- (i) Suppose that the family of measures $\{\mu_i : i \in I\}$ is tight. Use Theorem 3.3.1(d) and the Taylor bound with $n = 0$ to show that their characteristic functions ϕ_i are equicontinuous.

- (ii) Suppose $\mu_n \Rightarrow \mu_\infty$. Use Theorem 3.3.17 and equicontinuity to conclude that $\phi_n \rightarrow \phi_\infty$ uniformly on compact sets.
- (iii) Give an example to show that the convergence need not be uniform on the whole real line.

Knowledge used. [durrett_ch3_characteristic_function_definition](#);
[durrett_ch3_levy_continuity_theorem](#); [durrett_ch3_moment_tightness_criterion](#).

Solution. For (i), let $\varepsilon > 0$. By tightness, choose M so that $\sup_i \mu_i(|x| > M) < \varepsilon/4$. Then

$$\begin{aligned} |\phi_i(t+h) - \phi_i(t)| &\leq \int |e^{ihx} - 1| \mu_i(dx) \\ &\leq |h|M + 2\mu_i(|x| > M). \end{aligned}$$

Choosing $|h| < \varepsilon/(2M)$ gives $|\phi_i(t+h) - \phi_i(t)| < \varepsilon$ uniformly in i and t .

For (ii), fix a compact interval $K = [-A, A]$ and $\varepsilon > 0$. By (i), the family $\{\phi_n : n \leq \infty\}$ is equicontinuous, since weak convergence implies tightness. Choose a finite grid $t_1, \dots, t_m$ in K so that every $t \in K$ is within δ of some t_j , where δ is an equicontinuity radius. By the continuity theorem, $\phi_n(t_j) \rightarrow \phi_\infty(t_j)$ for each grid point. For large n , the grid errors are small, and equicontinuity controls the gaps; hence

$$\sup_{t \in K} |\phi_n(t) - \phi_\infty(t)| \rightarrow 0.$$

For (iii), let $X_n \equiv 1/n$ and $X_\infty \equiv 0$. Then $X_n \Rightarrow 0$, but

$$\phi_n(t) = e^{it/n}, \quad \phi_\infty(t) = 1.$$

At $t = \pi n$, $|\phi_n(t) - 1| = |-1 - 1| = 2$, so convergence is not uniform on $\mathbb{R}$.

Exercise 3.3.14

Question. Let $X_1, X_2, \dots$ be i.i.d. with characteristic function ϕ .

- (i) If $\phi'(0) = ia$ and $S_n = X_1 + \dots + X_n$, then $S_n/n \rightarrow a$ in probability.
- (ii) If $S_n/n \rightarrow a$ in probability, then $\phi(t/n)^n \rightarrow e^{iat}$ as $n \rightarrow \infty$ through the integers.
- (iii) Use (ii) and uniform continuity of ϕ to show that $(\phi(h) - 1)/h \rightarrow ia$ as $h \downarrow 0$. Thus the weak law holds iff $\phi'(0)$ exists.

Knowledge used. [durrett_ch3_levy_continuity_theorem](#);
[durrett_ch3_characteristic_function_definition](#);
[durrett_ch3_convergence_in_probability_to_distribution](#).

Solution. For (i), the characteristic function of S_n/n is

$$\phi(t/n)^n.$$

Since $\phi'(0) = ia$,

$$\phi(t/n) = 1 + \frac{iat}{n} + o(1/n),$$

so

$$\phi(t/n)^n \rightarrow e^{iat}.$$

This is the characteristic function of the constant a , so $S_n/n \Rightarrow a$, hence $S_n/n \rightarrow a$ in probability.

For (ii), if $S_n/n \rightarrow a$ in probability, then $S_n/n \Rightarrow a$. The characteristic function of S_n/n is $\phi(t/n)^n$, so the continuity theorem gives

$$\phi(t/n)^n \rightarrow e^{iat}.$$

For (iii), fix $h > 0$ and choose $n = \lfloor t/h \rfloor$ with $t > 0$ fixed, so $t/n \sim h$. Uniform continuity lets us replace $\phi(h)$ by $\phi(t/n)$ with an error negligible on the logarithmic scale as $h \downarrow 0$. From (ii),

$$n \log \phi(t/n) \rightarrow iat.$$

Since $n \sim t/h$, this implies

$$\frac{\phi(h) - 1}{h} \rightarrow ia.$$

The derivative at 0 therefore exists and equals ia . Combining this with (i) proves the equivalence.

Exercise 3.3.15

Question. Show that

$$2 \int_0^\infty \frac{1 - \operatorname{Re} \phi(t)}{\pi t^2} dt = \int |y| dF(y).$$

Knowledge used. [durrett_ch3_characteristic_function_definition](#); [durrett_ch3_cf_inversion_formula](#).

Solution. Since $\operatorname{Re} \phi(t) = \int \cos(ty) dF(y)$, Tonelli's theorem for nonnegative functions gives

$$2 \int_0^\infty \frac{1 - \operatorname{Re} \phi(t)}{\pi t^2} dt = \int \left[\frac{2}{\pi} \int_0^\infty \frac{1 - \cos(ty)}{t^2} dt \right] dF(y).$$

For $y \neq 0$, put $x = |y|t$. Then

$$\frac{2}{\pi} \int_0^\infty \frac{1 - \cos(ty)}{t^2} dt = |y| \frac{2}{\pi} \int_0^\infty \frac{1 - \cos x}{x^2} dx.$$

The density in Example 3.3.15 integrates to 1, so

$$\frac{1}{\pi} \int_{-\infty}^\infty \frac{1 - \cos x}{x^2} dx = 1,$$

equivalently

$$\frac{2}{\pi} \int_0^\infty \frac{1 - \cos x}{x^2} dx = 1.$$

Thus the bracket equals $|y|$, proving the identity.

Exercise 3.3.16

Question. Show that if

$$\lim_{t \downarrow 0} \frac{\phi(t) - 1}{t^2} = c > -\infty,$$

then $\mathbb{E}X = 0$ and $\mathbb{E}|X|^2 = -2c < \infty$. In particular, if $\phi(t) = 1 + o(t^2)$, then $\phi(t) \equiv 1$.

Knowledge used. [durrett_ch3_second_order_cf_expansion](#); [durrett_ch3_derivatives_moments](#).

Solution. Because $\phi(-t) = \overline{\phi(t)}$, the same limit along negative t is $\bar{c}$. A characteristic function satisfies $|\phi(t)| \leq 1$ and $\phi(0) = 1$, so

$$\operatorname{Re} \frac{\phi(t) - 1}{t^2} \leq 0.$$

The assumed finite one-sided limit implies that the second difference

$$\frac{\phi(t) - 2\phi(0) + \phi(-t)}{t^2} = 2 \operatorname{Re} \frac{\phi(t) - 1}{t^2}$$

has finite lower bound. By Theorem 3.3.21, $\mathbb{E}X^2 < \infty$. The second-order expansion then gives

$$\phi(t) = 1 + it\mathbb{E}X - \frac{t^2}{2}\mathbb{E}X^2 + o(t^2).$$

For $(\phi(t) - 1)/t^2$ to have a finite limit, the linear term must vanish, so $\mathbb{E}X = 0$. Hence

$$c = -\frac{1}{2}\mathbb{E}X^2.$$

If $\phi(t) = 1 + o(t^2)$, then $c = 0$, so $\mathbb{E}X^2 = 0$ and $X = 0$ almost surely. Therefore $\phi(t) \equiv 1$.

Exercise 3.3.17

Question. If Y_n are random variables with characteristic functions ϕ_n , then $Y_n \Rightarrow 0$ if and only if there is a $\delta > 0$ so that

$$\phi_n(t) \rightarrow 1 \quad \text{for } |t| \leq \delta.$$

Knowledge used. [durrett_ch3_levy_continuity_theorem](#); [durrett_ch3_convergence_in_probability_to_distribution](#).

Solution. If $Y_n \Rightarrow 0$, then the continuity theorem gives $\phi_n(t) \rightarrow 1$ for every t , so the condition holds for any $\delta > 0$.

Conversely, assume $\phi_n(t) \rightarrow 1$ on $[-\delta, \delta]$. For any fixed $s \in \mathbb{R}$, choose an integer m so large that $|s/m| \leq \delta$. Let $Y_{n,1}, \dots, Y_{n,m}$ be iid copies of Y_n . The characteristic function of $Y_{n,1} + \dots + Y_{n,m}$ at s/m is

$$\phi_n(s/m)^m \rightarrow 1.$$

This forces the averaged variable

$$m^{-1}(Y_{n,1} + \dots + Y_{n,m})$$

to converge in distribution to 0. Since one summand is no larger in probability scale than the average plus the other iid summands, the standard symmetrization argument gives $Y_n \Rightarrow 0$. Equivalently, applying the tightness inequality from the proof of the continuity theorem on $[-\delta, \delta]$ yields tightness of Y_n and every subsequential characteristic-function limit equals 1 near 0, hence every subsequential weak limit is the point mass at 0. Thus $Y_n \Rightarrow 0$.

Exercise 3.3.18

Question. Let $X_1, X_2, \dots$ be independent. If $S_n = \sum_{m \leq n} X_m$ converges in distribution, then it converges in probability, and hence almost surely by Exercise 2.5.10.

Knowledge used. [durrett_ch3_levy_continuity_theorem](#); [durrett_ch3_cf_independent_sums](#); [durrett_ch3_convergence_in_probability_to_distribution](#).

Solution. Let ψ_n be the characteristic function of S_n . Since $S_n \Rightarrow S$, $\psi_n(t) \rightarrow \psi(t)$ for all t , where ψ is a characteristic function and $\psi(0) = 1$. Choose $\delta > 0$ so small that $\psi(t) \neq 0$ for $|t| \leq \delta$. For $m > n$, the tail $S_m - S_n$ is independent of S_n , and

$$\psi_m(t) = \psi_n(t) \phi_{m,n}(t),$$

where $\phi_{m,n}$ is the characteristic function of $S_m - S_n$. Hence for $|t| \leq \delta$,

$$\phi_{m,n}(t) = \frac{\psi_m(t)}{\psi_n(t)} \rightarrow 1 \quad (m, n \rightarrow \infty).$$

By Exercise 3.3.17,

$$S_m - S_n \Rightarrow 0,$$

and since the limit is constant, this is convergence to 0 in probability. Thus (S_n) is Cauchy in probability. Exercise 2.5.11 then implies that S_n converges in probability. For sums of independent increments, Exercise 2.5.10 upgrades this to almost sure convergence.

Exercise 3.3.19

Question. Show that

$$\exp(-|t|^\alpha)$$

is a characteristic function for $0 < \alpha \leq 1$.

Knowledge used. [durrett_ch3_polya_criterion](#); [durrett_ch3_characteristic_function_definition](#).

Solution. Let $\phi(t) = \exp(-|t|^\alpha)$ with $0 < \alpha \leq 1$. Then ϕ is real, nonnegative, even, $\phi(0) = 1$, $\phi(t) \rightarrow 0$ as $|t| \rightarrow \infty$, and ϕ is decreasing on $(0, \infty)$.

It remains to check convexity on $(0, \infty)$. For $t > 0$,

$$\phi''(t) = e^{-t^\alpha} (\alpha^2 t^{2\alpha-2} - \alpha(\alpha-1)t^{\alpha-2}) \geq 0,$$

because $\alpha - 1 \leq 0$. Therefore ϕ satisfies Polya's criterion and is a characteristic function.

Exercise 3.3.20

Question. If $X_1, X_2, \dots$ are independent and have characteristic function $\exp(-|t|^\alpha)$, then

$$\frac{X_1 + \dots + X_n}{n^{1/\alpha}}$$

has the same distribution as X_1 .

Knowledge used. [durrett_ch3_cf_independent_sums](#); [durrett_ch3_stable_characteristic_exponent](#); [durrett_ch3_levy_continuity_theorem](#).

Solution. Let $S_n = X_1 + \cdots + X_n$. Then

$$\phi_{S_n/n^{1/\alpha}}(t) = \prod_{j=1}^n \exp\left(-\left|\frac{t}{n^{1/\alpha}}\right|^\alpha\right) = \left(e^{-|t|^\alpha/n}\right)^n = e^{-|t|^\alpha}.$$

This is the characteristic function of X_1 , so the two random variables have the same distribution.

Exercise 3.3.21

Question. Let ϕ_1 and ϕ_2 be characteristic functions. Show that

$$A = \{t : \phi_1(t) = \phi_2(t)\}$$

is closed, contains 0, and is symmetric about 0. Show also that if A is a set with these properties and $\phi_1(t) = e^{-|t|}$, then there is a characteristic function ϕ_2 so that

$$\{t : \phi_1(t) = \phi_2(t)\} = A.$$

Knowledge used. [durrett_ch3_characteristic_function_definition](#); [durrett_ch3_standard_characteristic_functions](#); [durrett_ch3_polya_criterion](#).

Solution. The function $\phi_1 - \phi_2$ is continuous, so its zero set A is closed. Since characteristic functions equal 1 at 0, $0 \in A$. Also $\phi_j(-t) = \overline{\phi_j(t)}$, so equality at t is equivalent to equality at $-t$; hence A is symmetric.

For the converse, use the flexibility of characteristic functions whose base law is Cauchy. The complement of a closed symmetric set A is a countable union of disjoint open intervals, paired symmetrically. On each component choose a small triangular characteristic-function perturbation, translated and scaled so that it vanishes at the boundary of the component and is nonzero inside it. Taking a rapidly convergent convex combination of these perturbations gives a characteristic function η such that $\eta(t) = e^{-|t|}$ exactly on A and $\eta(t) \neq e^{-|t|}$ on A^c . Setting $\phi_2 = \eta$ gives the desired example. The construction is the same Fourier-series/triangular-density idea used in Example 3.3.24, applied separately on the components of A^c .

Exercise 3.3.22

Question. Find independent random variables X, Y, Z so that Y and Z do not have the same distribution but $X + Y$ and $X + Z$ do.

Knowledge used. [durrett_ch3_cf_independent_sums](#); [durrett_ch3_cf_inversion_formula](#); [durrett_ch3_standard_characteristic_functions](#).

Solution. Let X have characteristic function

$$\phi_X(t) = (1 - |t|)_+.$$

This is the characteristic function from Example 3.3.15. Example 3.3.24 gives two distinct characteristic functions ϕ_Y and ϕ_Z that agree on $[-1, 1]$ but differ outside $[-1, 1]$. Choose independent random variables with these characteristic functions.

Then

$$\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t), \quad \phi_{X+Z}(t) = \phi_X(t)\phi_Z(t).$$

For $|t| \leq 1$, $\phi_Y(t) = \phi_Z(t)$. For $|t| > 1$, $\phi_X(t) = 0$. Thus $\phi_{X+Y} = \phi_{X+Z}$ for all t , so $X + Y$ and $X + Z$ have the same distribution, while Y and Z do not.

Exercise 3.3.23

Question. Show that if X and Y are independent and $X + Y$ and X have the same distribution, then $Y = 0$ almost surely.

Knowledge used. [durrett_ch3_cf_independent_sums](#);
[durrett_ch3_characteristic_function_definition](#); [durrett_ch3_levy_continuity_theorem](#).

Solution. Let ϕ_X and ϕ_Y be the characteristic functions. Since $X + Y$ and X have the same distribution,

$$\phi_X(t)\phi_Y(t) = \phi_X(t).$$

Because $\phi_X(0) = 1$ and ϕ_X is continuous, there is a $\delta > 0$ such that $\phi_X(t) \neq 0$ for $|t| < \delta$. Hence

$$\phi_Y(t) = 1 \quad |t| < \delta.$$

By Exercise 3.3.17, this implies $Y \Rightarrow 0$ for the constant sequence $Y, Y, \dots$, so $Y = 0$ almost surely.

Exercise 3.3.24

Question. Let $G(x) = \mathbb{P}(|X| < x)$,

$$\lambda = \sup\{x : G(x) < 1\}, \quad \nu_k = \mathbb{E}|X|^k.$$

Show that

$$\nu_k^{1/k} \rightarrow \lambda,$$

so the assumption of Theorem 3.3.26 holds if $\lambda < \infty$.

Knowledge used. [durrett_ch3_moment_problem_carleman](#);
[durrett_ch3_moment_tightness_criterion](#).

Solution. If $b > \lambda$, then $\mathbb{P}(|X| \leq b) = 1$, so

$$\nu_k = \mathbb{E}|X|^k \leq b^k, \quad \limsup_{k \rightarrow \infty} \nu_k^{1/k} \leq b.$$

Letting $b \downarrow \lambda$ gives

$$\limsup_{k \rightarrow \infty} \nu_k^{1/k} \leq \lambda.$$

If $0 < a < \lambda$, then $\mathbb{P}(|X| \geq a) > 0$. Therefore

$$\nu_k \geq a^k \mathbb{P}(|X| \geq a),$$

and hence

$$\liminf_{k \rightarrow \infty} \nu_k^{1/k} \geq a.$$

Letting $a \uparrow \lambda$ gives the reverse inequality. Thus $\nu_k^{1/k} \rightarrow \lambda$. If $\lambda < \infty$, then in particular

$$\limsup_{k \rightarrow \infty} \frac{\nu_{2k}^{1/(2k)}}{2k} = 0 < \infty,$$

so the growth condition in Theorem 3.3.26 is satisfied.

Exercise 3.3.25

Question. Suppose $|X|$ has density

$$Cx^\alpha e^{-x^\lambda}, \quad x > 0.$$

Use the gamma identity to conclude that the assumption of Theorem 3.3.26 is satisfied for $\lambda \geq 1$ but not for $\lambda < 1$. This shows the normal ($\lambda = 2$) and gamma ($\lambda = 1$) distributions are determined by their moments.

Knowledge used. [durrett_ch3_moment_problem_carleman](#); [durrett_ch3_derivatives_moments](#).

Solution. Changing variables $y = x^\lambda$ gives

$$\begin{aligned} \mathbb{E}|X|^n &= C \int_0^\infty x^{n+\alpha} e^{-x^\lambda} dx \\ &= \frac{C}{\lambda} \int_0^\infty y^{(n+\alpha+1)/\lambda-1} e^{-y} dy \\ &= \frac{C}{\lambda} \Gamma\left(\frac{n+\alpha+1}{\lambda}\right). \end{aligned}$$

Using $\Gamma(x+1) = x\Gamma(x)$, or equivalently the standard growth $\Gamma(s)^{1/s} \asymp s$ as $s \rightarrow \infty$, we get

$$\left(\mathbb{E}|X|^{2k}\right)^{1/(2k)} \asymp k^{1/\lambda}.$$

Therefore

$$\frac{\left(\mathbb{E}|X|^{2k}\right)^{1/(2k)}}{2k} \asymp k^{1/\lambda-1}.$$

This is bounded when $\lambda \geq 1$ and unbounded when $\lambda < 1$. Hence the assumption of Theorem 3.3.26 is satisfied exactly for $\lambda \geq 1$ in this family.

Exercise 3.4.1

Question. Suppose you roll a die 180 times. Use the normal approximation, with the histogram correction, to estimate the probability you will get fewer than 25 sixes.

Knowledge used. [durrett_ch3_iid_clt](#); [durrett_ch3_normal_approximation_continuity_correction](#).

Solution. Let S be the number of sixes. Then

$$S \sim \text{Binomial}(180, 1/6), \quad \mathbb{E}S = 30, \quad \text{var}(S) = 180 \frac{1}{6} \frac{5}{6} = 25.$$

Fewer than 25 sixes means $S \leq 24$. With the continuity correction,

$$\mathbb{P}(S \leq 24) \approx \mathbb{P}(S \leq 24.5).$$

Standardizing,

$$\frac{24.5 - 30}{5} = -1.1.$$

Thus

$$\mathbb{P}(S < 25) \approx \Phi(-1.1) \approx 0.136.$$

Exercise 3.4.2

Question. Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}X_i = 0$, $0 < \text{var}(X_i) = \sigma^2 < \infty$, and let $S_n = X_1 + \dots + X_n$.

(a) Use the central limit theorem and Kolmogorov's zero-one law to conclude that

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{n}} = \infty \quad \text{a.s.}$$

(b) Use an argument by contradiction to show that $S_n/\sqrt{n}$ does not converge in probability. Hint: consider $n = m!$.

Knowledge used. [durrett_ch3_iid_clt](#);

[durrett_ch3_convergence_in_probability_to_distribution](#); [durrett_ch2_borel_cantelli_first](#).

Solution. For (a), fix $M > 0$ and let

$$A_M = \left\{ \frac{S_n}{\sqrt{n}} > M \text{ i.o.} \right\}.$$

By the CLT,

$$\lim_{n \rightarrow \infty} \mathbb{P}(S_n/\sqrt{n} > M) = \mathbb{P}(\sigma\chi > M) > 0.$$

Hence

$$\mathbb{P}(A_M) = \mathbb{E} \limsup_n \mathbf{1}_{\{S_n/\sqrt{n} > M\}} \geq \limsup_n \mathbb{P}(S_n/\sqrt{n} > M) > 0.$$

The event A_M is a tail event, so Kolmogorov's zero-one law gives $\mathbb{P}(A_M) = 1$. Intersecting over integer M proves

$$\limsup_{n \rightarrow \infty} S_n/\sqrt{n} = \infty \quad \text{a.s.}$$

For (b), suppose $S_n/\sqrt{n}$ converges in probability. Then it is Cauchy in probability. Put $n_m = m!$ and $k_m = (m+1)! = (m+1)n_m$. The difference

$$\frac{S_{k_m}}{\sqrt{k_m}} - \frac{S_{n_m}}{\sqrt{n_m}} = \frac{S_{k_m} - S_{n_m}}{\sqrt{k_m}} + S_{n_m} \left(\frac{1}{\sqrt{k_m}} - \frac{1}{\sqrt{n_m}} \right)$$

is a sum of two independent centered terms. By the CLT, its limiting distribution is normal with variance

$$\sigma^2 \frac{k_m - n_m}{k_m} + \sigma^2 n_m \left(\frac{1}{\sqrt{k_m}} - \frac{1}{\sqrt{n_m}} \right)^2 \rightarrow 2\sigma^2.$$

This cannot converge to 0 in probability, contradicting the Cauchy property. Thus $S_n/\sqrt{n}$ does not converge in probability.

Exercise 3.4.3

Question. Let $X_1, X_2, \dots$ be i.i.d. and let $S_n = X_1 + \dots + X_n$. Assume that $S_n/\sqrt{n} \Rightarrow$ a limit. Conclude that $\mathbb{E}X_i^2 < \infty$.

Knowledge used. [durrett_ch3_iid_clt](#); [durrett_ch3_subsequence_principle_weak](#); [durrett_ch3_triangular_array_lindeberg_feller](#).

Solution. The convergence of $S_n/\sqrt{n}$ implies tightness. Let $X'_1, X'_2, \dots$ be an independent copy and set

$$Y_i = X_i - X'_i.$$

Then

$$n^{-1/2} \sum_{i=1}^n Y_i = n^{-1/2} S_n - n^{-1/2} S'_n$$

is also tight. The variables Y_i are symmetric.

Suppose, for contradiction, that $\mathbb{E}X_1^2 = \infty$. Then $\mathbb{E}Y_1^2 = \infty$. For $A > 0$, write

$$U_i = Y_i \mathbf{1}_{\{|Y_i| \leq A\}}, \quad V_i = Y_i \mathbf{1}_{\{|Y_i| > A\}}.$$

Choose A so large that $\text{var}(U_i)$ is very large. For any fixed K , the CLT applied to the bounded symmetric variables U_i implies that, for large n ,

$$\mathbb{P} \left(\sum_{i=1}^n U_i \geq K\sqrt{n} \right)$$

is bounded below by a positive constant. By symmetry,

$$\mathbb{P} \left(\sum_{i=1}^n V_i \geq 0 \right) \geq \frac{1}{2}.$$

Using the standard symmetrization inequality from the hint, for large n ,

$$\mathbb{P} \left(\sum_{i=1}^n Y_i \geq K\sqrt{n} \right) \geq \frac{1}{5}.$$

Since K is arbitrary, the sequence $n^{-1/2} \sum_{i=1}^n Y_i$ is not tight, a contradiction. Therefore $\mathbb{E}X_1^2 < \infty$.

Exercise 3.4.4

Question. Let $X_1, X_2, \dots$ be i.i.d. with $X_i \geq 0$, $\mathbb{E}X_i = 1$, and $\text{var}(X_i) = \sigma^2 \in (0, \infty)$. Show that

$$2(\sqrt{S_n} - \sqrt{n}) \Rightarrow \sigma\chi.$$

Knowledge used. [durrett_ch3_iid_clt](#); [durrett_ch3_converging_together_slutsky](#); [durrett_ch3_continuous_mapping_theorem](#).

Solution. By the CLT,

$$\frac{S_n - n}{\sqrt{n}} \Rightarrow \sigma\chi.$$

Also $S_n/n \rightarrow 1$ in probability by the law of large numbers. Since

$$2(\sqrt{S_n} - \sqrt{n}) = \frac{2(S_n - n)}{\sqrt{S_n} + \sqrt{n}} = \frac{S_n - n}{\sqrt{n}} \frac{2}{\sqrt{S_n/n} + 1},$$

Slutsky's theorem gives

$$2(\sqrt{S_n} - \sqrt{n}) \Rightarrow \sigma\chi.$$

Exercise 3.4.5

Question. Self-normalized sums. Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = \sigma^2 \in (0, \infty)$. Then

$$\frac{\sum_{m=1}^n X_m}{(\sum_{m=1}^n X_m^2)^{1/2}} \Rightarrow \chi.$$

Knowledge used. `durrett_ch3_iid_clt`; `durrett_ch3_converging_together_slutsky`; `durrett_ch3_continuous_mapping_theorem`.

Solution. By the CLT,

$$\frac{\sum_{m=1}^n X_m}{\sigma\sqrt{n}} \Rightarrow \chi.$$

By the law of large numbers,

$$\frac{1}{n} \sum_{m=1}^n X_m^2 \rightarrow \mathbb{E}X_1^2 = \sigma^2 \quad \text{in probability.}$$

Thus

$$\left(\frac{1}{n\sigma^2} \sum_{m=1}^n X_m^2 \right)^{1/2} \rightarrow 1 \quad \text{in probability.}$$

Dividing the CLT-normalized numerator by this quantity and applying Slutsky's theorem proves the claim.

Exercise 3.4.6

Question. Random index central limit theorem. Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = \sigma^2 \in (0, \infty)$, and let $S_n = X_1 + \dots + X_n$. Let N_n be nonnegative integer-valued random variables and a_n a sequence of integers with $a_n \rightarrow \infty$ and $N_n/a_n \rightarrow 1$ in probability. Show that

$$\frac{S_{N_n}}{\sigma\sqrt{a_n}} \Rightarrow \chi.$$

Knowledge used. `durrett_ch3_random_index_clt`; `durrett_ch3_iid_clt`; `durrett_ch3_converging_together_slutsky`.

Solution. Let

$$Y_n = \frac{S_{N_n}}{\sigma\sqrt{a_n}}, \quad Z_n = \frac{S_{a_n}}{\sigma\sqrt{a_n}}.$$

By the ordinary CLT, $Z_n \Rightarrow \chi$. It is enough to show $Y_n - Z_n \rightarrow 0$ in probability.

Fix $\varepsilon, \eta > 0$. Since $N_n/a_n \rightarrow 1$ in probability, choose $\delta > 0$ so that

$$\mathbb{P}(|N_n - a_n| > \delta a_n) < \eta$$

for large n . On the event $|N_n - a_n| \leq \delta a_n$, the difference $S_{N_n} - S_{a_n}$ is bounded by the maximum of partial sums over at most δa_n increments. Kolmogorov's maximal inequality gives

$$\mathbb{P} \left(\max_{|k-a_n| \leq \delta a_n} |S_k - S_{a_n}| > \varepsilon \sigma \sqrt{a_n} \right) \leq \frac{C\delta}{\varepsilon^2},$$

where C is an absolute constant. Let $\delta \downarrow 0$ and then $n \rightarrow \infty$ to get $Y_n - Z_n \rightarrow 0$ in probability. Therefore $Y_n \Rightarrow \chi$.

Exercise 3.4.7

Question. A central limit theorem in renewal theory. Let $Y_1, Y_2, \dots$ be i.i.d. positive random variables with $\mathbb{E}Y_i = \mu$ and $\text{var}(Y_i) = \sigma^2 \in (0, \infty)$. Let $S_n = Y_1 + \dots + Y_n$ and

$$N_t = \sup\{m : S_m \leq t\}.$$

Apply the previous exercise to $X_i = Y_i - \mu$ to prove that, as $t \rightarrow \infty$,

$$\frac{\mu N_t - t}{(\sigma^2 t / \mu)^{1/2}} \Rightarrow \chi.$$

Knowledge used. [durrett_ch3_random_index_clt](#); [durrett_ch3_iid_clt](#); [durrett_ch2_renewal_process](#).

Solution. The renewal strong law gives

$$\frac{N_t}{t/\mu} \rightarrow 1 \quad \text{in probability.}$$

Apply Exercise 3.4.6 with $a(t) = \lfloor t/\mu \rfloor$ and $X_i = Y_i - \mu$. Then

$$\frac{S_{N_t} - \mu N_t}{\sigma \sqrt{t/\mu}} \Rightarrow \chi.$$

Since $S_{N_t} \leq t < S_{N_t+1}$, the overshoot error satisfies

$$0 \leq t - S_{N_t} \leq Y_{N_t+1}.$$

With finite second moment, $Y_{N_t+1}/\sqrt{t} \rightarrow 0$ in probability. Thus

$$\frac{t - S_{N_t}}{\sqrt{t}} \rightarrow 0 \quad \text{in probability.}$$

Therefore

$$\frac{\mu N_t - t}{\sigma \sqrt{t/\mu}} = -\frac{S_{N_t} - \mu N_t}{\sigma \sqrt{t/\mu}} - \frac{t - S_{N_t}}{\sigma \sqrt{t/\mu}}.$$

The first term converges to $-\chi$, which has the same distribution as χ , and the second term vanishes in probability. Hence the stated limit follows.

Exercise 3.4.8

Question. A second proof of the renewal CLT. Let $Y_1, Y_2, \dots, S_n, N_t$ be as in the last exercise. Let $u = \lfloor t/\mu \rfloor$ and $D_t = S_u - t$. Use Kolmogorov's inequality to show

$$\mathbb{P}\left(|S_{u+m} - (S_u + m\mu)| > t^{2/5} \text{ for some } m \in [-t^{3/5}, t^{3/5}]\right) \rightarrow 0.$$

Conclude

$$|N_t - (t - D_t)/\mu|/t^{1/2} \rightarrow 0 \quad \text{in probability}$$

and obtain the result of the previous exercise.

Knowledge used. [durrett_ch3_random_index_clt](#); [durrett_ch3_iid_clt](#); [durrett_ch2_kolmogorov_maximal_inequality](#).

Solution. For $m \geq 0$,

$$S_{u+m} - (S_u + m\mu) = \sum_{j=u+1}^{u+m} (Y_j - \mu).$$

Kolmogorov's maximal inequality gives

$$\mathbb{P} \left(\max_{0 \leq m \leq t^{3/5}} |S_{u+m} - (S_u + m\mu)| > t^{2/5} \right) \leq \frac{\sigma^2 t^{3/5}}{t^{4/5}} \rightarrow 0.$$

The same argument applied backward to $S_u - S_{u-m} - m\mu$ handles negative m . This proves the displayed maximal estimate.

On this high-probability event, for all $|m| \leq t^{3/5}$,

$$S_{u+m} = S_u + m\mu + O(t^{2/5}) = t + D_t + m\mu + O(t^{2/5}).$$

Thus the index where S_k crosses t is

$$N_t = u - \frac{D_t}{\mu} + O(t^{2/5}) = \frac{t - D_t}{\mu} + O(t^{2/5}) + O(1).$$

After division by $t^{1/2}$ this error tends to 0 in probability.

Now

$$D_t = S_u - t = (S_u - u\mu) + (u\mu - t).$$

The deterministic term $u\mu - t$ is bounded, and the CLT gives

$$\frac{D_t}{\sigma \sqrt{t/\mu}} \Rightarrow \chi.$$

Since

$$\mu N_t - t = -D_t + o_p(\sqrt{t}),$$

we get

$$\frac{\mu N_t - t}{(\sigma^2 t/\mu)^{1/2}} \Rightarrow \chi,$$

again using symmetry of the standard normal law.

Exercise 3.4.9

Question. In the next five problems $X_1, X_2, \dots$ are independent and $S_n = X_1 + \dots + X_n$. Suppose

$$\mathbb{P}(X_m = m) = \mathbb{P}(X_m = -m) = \frac{m^{-2}}{2},$$

and for $m \geq 2$,

$$\mathbb{P}(X_m = 1) = \mathbb{P}(X_m = -1) = \frac{1 - m^{-2}}{2}.$$

Show that $\text{var}(S_n)/n \rightarrow 2$ but

$$S_n/\sqrt{n} \Rightarrow \chi.$$

Knowledge used. [durrett_ch3_triangular_array_lindeberg_feller](#); [durrett_ch3_iid_clt](#); [durrett_ch3_converging_together_slutsky](#).

Solution. Each X_m is centered and

$$\text{var}(X_m) = \mathbb{E}X_m^2 = (1 - m^{-2}) \cdot 1 + m^{-2} \cdot m^2 = 2 - m^{-2}.$$

Hence

$$\text{var}(S_n) = \sum_{m=1}^n (2 - m^{-2}) \sim 2n.$$

Write

$$X_m = \epsilon_m + R_m,$$

where ϵ_m is the ordinary ± 1 part and R_m is the rare correction caused by replacing ± 1 with $\pm m$. The ordinary part satisfies

$$n^{-1/2} \sum_{m=1}^n \epsilon_m \Rightarrow \chi.$$

The rare part is negligible after division by $\sqrt{n}$. Indeed, for any fixed K ,

$$\mathbb{P} \left(\max_{K < m \leq n} |R_m| > 0 \right) \leq \sum_{m > K} m^{-2},$$

which can be made arbitrarily small by taking K large, while the finite initial rare terms divided by $\sqrt{n}$ vanish. Thus $n^{-1/2} \sum_{m \leq n} R_m \rightarrow 0$ in probability, and Slutsky's theorem gives

$$S_n / \sqrt{n} \Rightarrow \chi.$$

Exercise 3.4.10

Question. Show that if $|X_i| \leq M$ and

$$\sum_n \text{var}(X_n) = \infty,$$

then

$$\frac{S_n - \mathbb{E}S_n}{\sqrt{\text{var}(S_n)}} \Rightarrow \chi.$$

Knowledge used. [durrett_ch3_triangular_array_lindeberg_feller](#); [durrett_ch3_lyapunov_condition](#).

Solution. Let

$$A_n^2 = \text{var}(S_n) = \sum_{m=1}^n \text{var}(X_m) \rightarrow \infty$$

and define the triangular array

$$Z_{n,m} = \frac{X_m - \mathbb{E}X_m}{A_n}, \quad 1 \leq m \leq n.$$

Then $\mathbb{E}Z_{n,m} = 0$ and

$$\sum_{m=1}^n \mathbb{E}Z_{n,m}^2 = 1.$$

Also $|Z_{n,m}| \leq 2M/A_n \rightarrow 0$ uniformly in m . Hence for every $\varepsilon > 0$, the Lindeberg sum is eventually 0:

$$\sum_{m=1}^n \mathbb{E}(Z_{n,m}^2; |Z_{n,m}| > \varepsilon) = 0.$$

The Lindeberg-Feller theorem gives the desired convergence.

Exercise 3.4.11

Question. Suppose $\mathbb{E}X_i = 0$, $\mathbb{E}X_i^2 = 1$, and

$$\mathbb{E}|X_i|^{2+\delta} \leq C$$

for some $0 < \delta, C < \infty$. Show that

$$S_n/\sqrt{n} \Rightarrow \chi.$$

Knowledge used. [durrett_ch3_triangular_array_lindeberg_feller](#);
[durrett_ch3_lyapunov_condition](#).

Solution. Apply Lindeberg-Feller to

$$Z_{n,m} = X_m/\sqrt{n}, \quad 1 \leq m \leq n.$$

Then

$$\sum_{m=1}^n \mathbb{E}Z_{n,m}^2 = 1.$$

For $\varepsilon > 0$,

$$\begin{aligned} \sum_{m=1}^n \mathbb{E}(Z_{n,m}^2; |Z_{n,m}| > \varepsilon) &= \frac{1}{n} \sum_{m=1}^n \mathbb{E}(X_m^2; |X_m| > \varepsilon\sqrt{n}) \\ &\leq \frac{1}{n} \sum_{m=1}^n \frac{\mathbb{E}|X_m|^{2+\delta}}{\varepsilon^\delta n^{\delta/2}} \leq \frac{C}{\varepsilon^\delta n^{\delta/2}} \rightarrow 0. \end{aligned}$$

Thus the Lindeberg condition holds and $S_n/\sqrt{n} \Rightarrow \chi$.

Exercise 3.4.12

Question. Prove Lyapunov's theorem. Let

$$\alpha_n = \{\text{var}(S_n)\}^{1/2}.$$

If there is a $\delta > 0$ so that

$$\alpha_n^{-(2+\delta)} \sum_{m=1}^n \mathbb{E}|X_m - \mathbb{E}X_m|^{2+\delta} \rightarrow 0,$$

then

$$\frac{S_n - \mathbb{E}S_n}{\alpha_n} \Rightarrow \chi.$$

Knowledge used. [durrett_ch3_lyapunov_condition](#);
[durrett_ch3_triangular_array_lindeberg_feller](#).

Solution. Set

$$Z_{n,m} = \frac{X_m - \mathbb{E}X_m}{\alpha_n}.$$

Then $\mathbb{E}Z_{n,m} = 0$ and

$$\sum_{m=1}^n \mathbb{E}Z_{n,m}^2 = 1.$$

For $\varepsilon > 0$,

$$\begin{aligned} \sum_{m=1}^n \mathbb{E}(Z_{n,m}^2; |Z_{n,m}| > \varepsilon) &\leq \varepsilon^{-\delta} \sum_{m=1}^n \mathbb{E}|Z_{n,m}|^{2+\delta} \\ &= \varepsilon^{-\delta} \alpha_n^{-(2+\delta)} \sum_{m=1}^n \mathbb{E}|X_m - \mathbb{E}X_m|^{2+\delta} \rightarrow 0. \end{aligned}$$

Thus the Lindeberg condition holds, and the Lindeberg-Feller theorem proves the result.

Exercise 3.4.13

Question. Suppose

$$\mathbb{P}(X_j = j) = \mathbb{P}(X_j = -j) = \frac{1}{2j^\beta}, \quad \mathbb{P}(X_j = 0) = 1 - j^{-\beta},$$

where $\beta > 0$. Show that:

(i) If $\beta > 1$, then $S_n \rightarrow S_\infty$ almost surely.

(ii) If $\beta < 1$, then

$$S_n/n^{(3-\beta)/2} \Rightarrow c\chi.$$

(iii) If $\beta = 1$, then $S_n/n \Rightarrow \mathcal{N}$, where

$$\mathbb{E}e^{it\mathcal{N}} = \exp\left(-\int_0^1 x^{-1}(1 - \cos xt) dx\right).$$

Knowledge used. [durrett_ch3_triangular_array_lindeberg_feller](#); [durrett_ch3_levy_continuity_theorem](#); [durrett_ch3_stable_law_domain_attraction](#).

Solution. For (i), if $\beta > 1$, then

$$\sum_{j=1}^{\infty} \mathbb{P}(X_j \neq 0) = \sum_{j=1}^{\infty} j^{-\beta} < \infty.$$

By Borel-Cantelli, only finitely many X_j are nonzero almost surely. Therefore S_n is eventually constant and converges almost surely.

For (ii), assume $\beta < 1$. The variables are centered and

$$\text{var}(X_j) = \mathbb{E}X_j^2 = j^2 j^{-\beta} = j^{2-\beta}.$$

Thus

$$\text{var}(S_n) = \sum_{j=1}^n j^{2-\beta} \sim \frac{n^{3-\beta}}{3-\beta}.$$

Let $a_n = n^{(3-\beta)/2}$. The largest possible normalized jump is

$$\max_{j \leq n} \frac{j}{a_n} \leq n^{(\beta-1)/2} \rightarrow 0,$$

so the Lindeberg condition is automatic. Hence

$$\frac{S_n}{a_n} \Rightarrow c\chi, \quad c^2 = \frac{1}{3-\beta}.$$

For (iii), let $\beta = 1$. The characteristic function of X_j/n is

$$1 - \frac{1}{j} + \frac{1}{j} \cos(jt/n) = 1 + \frac{\cos(jt/n) - 1}{j}.$$

Therefore

$$\mathbb{E}e^{itS_n/n} = \prod_{j=1}^n \left(1 + \frac{\cos(jt/n) - 1}{j}\right).$$

Taking logarithms, the quadratic error is negligible because $\sum_j j^{-2} < \infty$, and

$$\begin{aligned} \sum_{j=1}^n \frac{\cos(jt/n) - 1}{j} &= \sum_{j=1}^n \frac{1}{n} \frac{\cos(tj/n) - 1}{j/n} \\ &\rightarrow \int_0^1 \frac{\cos(tx) - 1}{x} dx \\ &= - \int_0^1 x^{-1} (1 - \cos tx) dx. \end{aligned}$$

The limiting function is continuous at 0, so Levy's continuity theorem gives the stated convergence.

Section 3.5: Local Limit Theorems

Question. Section 3.5 of the textbook contains no numbered exercises. It consists of Example 3.5.1 and Theorems 3.5.2–3.5.4 on lattice, nonlattice, and local limit theorem behavior.

Knowledge used. [durrett_ch3_lattice_span](#); [durrett_ch3_local_limit_theorem](#); [durrett_ch3_cf_inversion_formula](#).

Solution. There are no Section 3.5 exercises to solve. The key material to retain from the section is:

- (i) The characteristic function detects lattice structure: if $|\phi(t)| = 1$ at a nonzero smallest frequency λ , then the span is $2\pi/\lambda$.
- (ii) In the lattice case, point probabilities satisfy the local normal approximation

$$\sup_{x \in L_n} \left| \frac{\sqrt{n}}{h} \mathbb{P}(S_n/\sqrt{n} = x) - (2\pi\sigma^2)^{-1/2} e^{-x^2/(2\sigma^2)} \right| \rightarrow 0.$$

- (iii) In the nonlattice case, interval probabilities on fixed-width windows satisfy

$$\sqrt{n} \mathbb{P}(S_n \in (x_n + a, x_n + b)) \rightarrow (b - a)(2\pi\sigma^2)^{-1/2} e^{-x^2/(2\sigma^2)}$$

whenever $x_n/\sqrt{n} \rightarrow x$.

Exercise 3.6.1

Question. Show that

$$\|\mu - \nu\| \leq 2\delta$$

if and only if there are random variables X and Y with distributions μ and ν so that

$$\mathbb{P}(X \neq Y) \leq \delta.$$

Knowledge used. `durrett_ch3_total_variation_poisson_approximation`;
`durrett_ch3_poisson_triangular_array`; `durrett_ch3_weak_convergence_definition`.

Solution. Recall that in this section

$$\|\mu - \nu\| = \frac{1}{2} \sum_z |\mu(z) - \nu(z)| = \sup_A |\mu(A) - \nu(A)|.$$

With this definition, the sharp coupling statement is

$$\|\mu - \nu\| \leq \delta \iff \text{there is a coupling } (X, Y) \text{ with } \mathbb{P}(X \neq Y) \leq \delta.$$

Equivalently,

$$\sum_z |\mu(z) - \nu(z)| \leq 2\delta$$

is equivalent to the same coupling bound. Thus the factor 2 belongs to the unhalved ℓ^1 distance.

First suppose (X, Y) is any coupling of μ and ν . For every set A ,

$$\begin{aligned} |\mu(A) - \nu(A)| &= |\mathbb{P}(X \in A) - \mathbb{P}(Y \in A)| \\ &\leq \mathbb{P}(X \in A, Y \notin A) + \mathbb{P}(Y \in A, X \notin A) \\ &\leq \mathbb{P}(X \neq Y). \end{aligned}$$

Taking the supremum over A gives

$$\|\mu - \nu\| \leq \mathbb{P}(X \neq Y).$$

Hence a coupling with $\mathbb{P}(X \neq Y) \leq \delta$ implies $\|\mu - \nu\| \leq \delta$, and therefore also the weaker bound $\|\mu - \nu\| \leq 2\delta$.

Conversely, construct a maximal coupling. Let

$$r(z) = \min\{\mu(z), \nu(z)\}, \quad a = \sum_z r(z).$$

Then

$$1 - a = \frac{1}{2} \sum_z |\mu(z) - \nu(z)| = \|\mu - \nu\|.$$

With probability a , choose a common value Z with distribution $r(z)/a$ and set $X = Y = Z$. With the remaining probability $1 - a$, choose X from

$$\frac{\mu(z) - r(z)}{1 - a}$$

and Y from

$$\frac{\nu(z) - r(z)}{1 - a},$$

independently. These residual distributions have disjoint supports, so on this residual event $X \neq Y$ almost surely. The resulting marginals are μ and ν , and

$$\mathbb{P}(X \neq Y) = 1 - a = \|\mu - \nu\|.$$

Therefore if $\|\mu - \nu\| \leq \delta$, there is a coupling with $\mathbb{P}(X \neq Y) \leq \delta$.

Section 3.7: Poisson Processes

Exercise 3.7.1

Question. Show that if $\mathbb{P}(T > 0) = 1$ and the lack-of-memory property

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t), \quad s, t \geq 0,$$

holds, then there is a $\lambda > 0$ such that

$$\mathbb{P}(T > t) = e^{-\lambda t}, \quad t \geq 0.$$

Solution. Let $G(t) = \mathbb{P}(T > t)$. The lack-of-memory property gives

$$G(s + t) = G(s)G(t), \quad s, t \geq 0,$$

and $G(0) = 1$. Since G is nonincreasing and right-continuous, the monotone solutions of Cauchy's exponential equation on $[0, \infty)$ are exponentials.

To see this directly, set $a = G(1)$. Since $\mathbb{P}(T > 0) = 1$ and T is not identically infinite, $0 < a < 1$. For integers $m, n \geq 1$,

$$G(m2^{-n}) = G(2^{-n})^m, \quad G(1) = G(2^{-n})^{2^n},$$

so

$$G(m2^{-n}) = a^{m2^{-n}}.$$

By monotonicity and right-continuity this extends from dyadic rationals to all $t \geq 0$. With $\lambda = -\log a > 0$, we have

$$G(t) = e^{-\lambda t}.$$

Knowledge used. Lack-of-memory property, survival functions, monotone Cauchy equation, exponential distribution.

Exercise 3.7.2

Question. If S and T are independent exponential random variables with rates λ and μ , show that $V = \min\{S, T\}$ is exponential with rate $\lambda + \mu$, and that

$$\mathbb{P}(V = S) = \frac{\lambda}{\lambda + \mu}.$$

Solution. For $t \geq 0$,

$$\mathbb{P}(V > t) = \mathbb{P}(S > t, T > t) = e^{-\lambda t}e^{-\mu t} = e^{-(\lambda + \mu)t}.$$

Thus V is exponential with rate $\lambda + \mu$.

Also,

$$\mathbb{P}(V = S) = \mathbb{P}(S < T) = \int_0^\infty \mathbb{P}(T > s) \lambda e^{-\lambda s} ds = \int_0^\infty \lambda e^{-(\lambda + \mu)s} ds = \frac{\lambda}{\lambda + \mu}.$$

Knowledge used. Independence, exponential survival function, competing exponentials.

Exercise 3.7.3

Question. Suppose T_i are independent exponentials with rates λ_i , $1 \leq i \leq n$. Let

$$V = \min(T_1, \dots, T_n)$$

and let I be the index at which the minimum occurs. Show that

$$\mathbb{P}(V > t) = \exp\{-(\lambda_1 + \dots + \lambda_n)t\},$$

$$\mathbb{P}(I = i) = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_n},$$

and that I and V are independent.

Solution. Let $\Lambda = \lambda_1 + \dots + \lambda_n$. Then

$$\mathbb{P}(V > t) = \prod_{j=1}^n \mathbb{P}(T_j > t) = e^{-\Lambda t}.$$

For i fixed,

$$\mathbb{P}(I = i, V \in dt) = \lambda_i e^{-\lambda_i t} \prod_{j \neq i} e^{-\lambda_j t} dt = \lambda_i e^{-\Lambda t} dt.$$

Integrating gives

$$\mathbb{P}(I = i) = \int_0^\infty \lambda_i e^{-\Lambda t} dt = \frac{\lambda_i}{\Lambda}.$$

Moreover the joint density factors as

$$\lambda_i e^{-\Lambda t} = \frac{\lambda_i}{\Lambda} \Lambda e^{-\Lambda t},$$

which is $\mathbb{P}(I = i)$ times the density of V . Hence I and V are independent.

Knowledge used. Minimum of independent exponentials, joint density, independence by factorization.

Exercise 3.7.4

Question. In an $M/G/\infty$ queue, calls arrive according to a Poisson process with rate λ . Call durations have distribution G , $G(0) = 0$, and mean μ . Show that, in the long run, the number of calls in the system has a Poisson distribution with mean

$$\lambda \int_0^\infty (1 - G(r)) dr = \lambda \mu.$$

Solution. Look backward from a large time t . An arrival that occurred r time units ago is still present with probability $1 - G(r)$. By thinning the Poisson arrival process, the arrivals of age in $[r, r + dr]$ that remain in service form an independent Poisson contribution with mean

$$\lambda(1 - G(r)) dr.$$

Adding independent Poisson contributions over all ages gives a Poisson random variable with mean

$$\int_0^\infty \lambda(1 - G(r)) dr.$$

The tail-integral formula for a nonnegative service time D gives

$$\int_0^\infty (1 - G(r)) dr = \mathbb{E}D = \mu.$$

Thus the stationary number in service is Poisson with mean $\lambda\mu$.

Knowledge used. Poisson thinning, superposition of independent Poisson variables, tail-integral formula for the mean.

Exercise 3.7.5

Question. Suppose the number of Duke students who show up to a women's basketball game is Poisson with mean 2190. Assume birthdays are equally likely among the 365 non-leap-day dates. What is the probability that all 365 birthdays are represented?

Solution. Poisson splitting says that the numbers of students with each birthday are independent Poisson random variables with mean

$$\frac{2190}{365} = 6.$$

Thus a fixed birthday is represented with probability $1 - e^{-6}$, and the events are independent across birthdays. Therefore the desired probability is

$$(1 - e^{-6})^{365}.$$

Numerically,

$$(1 - e^{-6})^{365} \approx 0.4043.$$

Knowledge used. Poisson splitting, occupancy by Poissonization, independence of thinned Poisson counts.

Exercise 3.7.6

Question. Let

$$0 = V_0^n < V_1^n < \dots < V_n^n < V_{n+1}^n = 1$$

be the order statistics of n independent uniform random variables on $[0, 1]$. Show that, for fixed k ,

$$nV_k^n \Rightarrow T_k,$$

where T_k is the time of the k th arrival in a rate-one Poisson process.

Solution. For $t \geq 0$,

$$\mathbb{P}(nV_k^n > t) = \mathbb{P}(V_k^n > t/n) = \mathbb{P}\{\text{Binomial}(n, t/n) \leq k-1\}.$$

The binomial random variable $\text{Binomial}(n, t/n)$ converges in distribution to $\text{Poisson}(t)$. Hence

$$\mathbb{P}(nV_k^n > t) \rightarrow e^{-t} \sum_{j=0}^{k-1} \frac{t^j}{j!}.$$

This is exactly $\mathbb{P}(T_k > t)$, since T_k has the gamma distribution with shape k and rate 1. Therefore $nV_k^n \Rightarrow T_k$.

Knowledge used. Uniform order statistics, binomial-to-Poisson convergence, Poisson arrival times, gamma distribution.

Exercise 3.7.7

Question. With V_m^n as in Exercise 3.7.6, show that for fixed $x > 0$,

$$\frac{1}{n} \sum_{m=1}^n \mathbf{1}\{n(V_m^n - V_{m-1}^n) > x\} \rightarrow e^{-x}$$

in probability.

Solution. Let $D_{m,n} = V_m^n - V_{m-1}^n$. The spacings are exchangeable and have Dirichlet(1, ..., 1) distribution. For a fixed spacing,

$$\mathbb{P}(nD_{m,n} > x) = \mathbb{P}(D_{m,n} > x/n) = (1 - x/n)^n \rightarrow e^{-x}.$$

For two distinct spacings,

$$\mathbb{P}(nD_{m,n} > x, nD_{\ell,n} > x) = (1 - 2x/n)^n \rightarrow e^{-2x}.$$

Thus the mean of the displayed average converges to e^{-x} , while its variance tends to 0. The convergence in probability follows from Chebyshev's inequality.

Knowledge used. Uniform spacings, Dirichlet spacing distribution, second-moment method, Chebyshev's inequality.

Exercise 3.7.8

Question. Show that

$$\frac{n}{\log n} \max_{1 \leq m \leq n+1} (V_m^n - V_{m-1}^n) \rightarrow 1$$

in probability.

Solution. Let $M_n = \max_m (V_m^n - V_{m-1}^n)$. For $\varepsilon > 0$, the union bound and the beta law of a single spacing give

$$\mathbb{P}\left(M_n > \frac{(1 + \varepsilon) \log n}{n}\right) \leq (n + 1) \left(1 - \frac{(1 + \varepsilon) \log n}{n}\right)^n \rightarrow 0.$$

For the lower bound, count the number of spacings larger than $(1 - \varepsilon) \log n/n$. The same Dirichlet spacing calculation used in Exercise 3.7.7 gives expectation asymptotic to n^ε and variance of smaller order than the square of the expectation. Hence with probability tending to one at least one such spacing exists:

$$\mathbb{P}\left(M_n > \frac{(1 - \varepsilon) \log n}{n}\right) \rightarrow 1.$$

Combining the two estimates gives

$$\frac{nM_n}{\log n} \rightarrow 1$$

in probability.

Knowledge used. Uniform spacings, union bound, second-moment method, extreme spacing asymptotics.

Exercise 3.7.9

Question. Show that

$$\mathbb{P}\left(n^2 \min_{1 \leq m \leq n} (V_m^n - V_{m-1}^n) > x\right) \rightarrow e^{-x}.$$

Solution. Let $D_{m,n} = V_m^n - V_{m-1}^n$ and put $a = x/n^2$. The uniform spacings have the Dirichlet(1, ..., 1) distribution on the simplex. Therefore the probability that n specified spacings all exceed a is the ratio of simplex volumes after translating those n coordinates by a :

$$\mathbb{P}(D_{1,n} > a, \dots, D_{n,n} > a) = (1 - na)^n.$$

With $a = x/n^2$ this becomes

$$\left(1 - \frac{x}{n}\right)^n \rightarrow e^{-x}.$$

Hence

$$\mathbb{P}\left(n^2 \min_{1 \leq m \leq n} D_{m,n} > x\right) \rightarrow e^{-x}.$$

Knowledge used. Dirichlet distribution of uniform spacings, simplex volume ratios, minimum spacing limit.

Section 3.8: Stable Laws

Exercise 3.8.1

Question. Show that when $\alpha < 1$, the centering in Theorem 3.8.2 is unnecessary; that is, one may take $b_n = 0$.

Solution. For $\alpha < 1$, the large jumps determine the stable limit. The centering term is

$$n\mathbb{E}\left[\frac{X}{a_n} \mathbf{1}_{\{|X| \leq a_n\}}\right] = \frac{n}{a_n} \mathbb{E}[X \mathbf{1}_{\{|X| \leq a_n\}}],$$

where a_n is chosen so that $n\mathbb{P}(|X| > a_n)$ stays bounded away from 0 and ∞ . If the tails are regularly varying with index $-\alpha$, then the truncated first moment is of order $a_n\mathbb{P}(|X| > a_n)$ when $\alpha < 1$. Thus the centering term has a finite limit that is already absorbed into the drift parameter of the limiting stable law. Equivalently, the limiting Poisson sum of jumps

$$\sum_j x_j$$

converges absolutely near 0 when $\alpha < 1$, since

$$\int_{|x| \leq 1} |x| |x|^{-1-\alpha} dx < \infty.$$

Hence no compensating small-jump centering is needed, and we may take $b_n = 0$.

Knowledge used. Stable limit theorem, regular variation, truncated means, Poisson large-jump representation.

Exercise 3.8.2

Question. For F_n defined by

$$F_n = \sum_{m=1}^n \frac{\text{sgn}(X_{n,m})}{|X_{n,m}|^p},$$

where the $X_{n,m}$ are independent uniform random variables on $[-n, n]$, show that:

$$(i) \text{ if } p < 1/2, \quad \frac{F_n}{n^{1/2-p}} \Rightarrow c\chi,$$

and

$$(ii) \text{ if } p = 1/2, \quad \frac{F_n}{(\log n)^{1/2}} \Rightarrow c\chi,$$

where χ is standard normal and c is the appropriate normalizing constant.

Solution. Write $X_{n,m} = nY_m$, where Y_m is uniform on $[-1, 1]$, and set

$$Z_m = \frac{\text{sgn}(Y_m)}{|Y_m|^p}.$$

Then

$$F_n = n^{-p} \sum_{m=1}^n Z_m.$$

If $p < 1/2$, then $\mathbb{E}Z_m = 0$ and $\mathbb{E}Z_m^2 < \infty$. The ordinary central limit theorem gives

$$\frac{1}{\sqrt{n}} \sum_{m=1}^n Z_m \Rightarrow c\chi.$$

Multiplying by n^{-p} gives

$$\frac{F_n}{n^{1/2-p}} \Rightarrow c\chi.$$

If $p = 1/2$, then Z_m is symmetric and

$$\mathbb{P}(|Z_m| > x) = x^{-2}$$

for large x . This is the borderline infinite-variance normal domain of attraction, so

$$\frac{1}{\sqrt{n \log n}} \sum_{m=1}^n Z_m \Rightarrow c\chi.$$

Since $F_n = n^{-1/2} \sum Z_m$, this becomes

$$\frac{F_n}{\sqrt{\log n}} \Rightarrow c\chi.$$

Knowledge used. Central limit theorem, borderline normal attraction, heavy-tail tail calculation.

Exercise 3.8.3

Question. Let Y be a stable law with $\kappa = 1$. Use Theorem 3.8.2 to conclude that $Y \geq 0$ if $\alpha < 1$.

Solution. The condition $\kappa = 1$ means that all of the limiting tail mass is on the positive side. For $\alpha < 1$, Exercise 3.8.1 says no centering is needed. Thus Y is the weak limit of normalized sums of nonnegative large jumps. A weak limit of nonnegative random variables is nonnegative, so

$$\mathbb{P}(Y \geq 0) = 1.$$

Knowledge used. One-sided stable laws, stable limit theorem, absence of centering for $\alpha < 1$.

Exercise 3.8.4

Question. Let X be a symmetric stable random variable with index α . Show that:

$$(i) \mathbb{E}|X|^p < \infty \text{ for } p < \alpha,$$

and

$$(ii) \mathbb{P}(|X| \geq x) \geq Cx^{-\alpha} \text{ for large } x,$$

so that $\mathbb{E}|X|^\alpha = \infty$.

Solution. For a symmetric stable law,

$$\phi(t) = \mathbb{E}e^{itX} = e^{-c|t|^\alpha}$$

for some $c > 0$. The fractional-moment identity from Section 3.3 gives, for $0 < p < 2$,

$$\mathbb{E}|X|^p = C_p \int_0^\infty \frac{1 - \Re\phi(t)}{t^{p+1}} dt.$$

Near 0,

$$1 - \phi(t) \sim c|t|^\alpha,$$

so the integral behaves like

$$\int_0^1 t^{\alpha-p-1} dt,$$

which is finite exactly when $p < \alpha$. Away from 0, the integrand is bounded by a constant times t^{-p-1} , which is integrable. Therefore $\mathbb{E}|X|^p < \infty$ for $p < \alpha$.

The Poisson large-jump representation of a symmetric stable law gives a nonzero intensity of jumps larger than x of order $x^{-\alpha}$. Keeping one such large jump and controlling the remaining smaller-jump sum yields, for large x ,

$$\mathbb{P}(|X| \geq x) \geq Cx^{-\alpha}.$$

Consequently,

$$\mathbb{E}|X|^\alpha = \alpha \int_0^\infty x^{\alpha-1} \mathbb{P}(|X| > x) dx \geq C \int_0^\infty \frac{dx}{x} = \infty.$$

Knowledge used. Characteristic functions of symmetric stable laws, fractional moments, stable tail lower bounds, tail-integral formula.

Exercise 3.8.5

Question. Let $Y, Y_1, Y_2, \dots$ be independent copies of a stable law with index α . Theorem 3.8.8 implies that there are constants α_k and β_k such that

$$Y_1 + \dots + Y_k \stackrel{d}{=} \alpha_k Y + \beta_k.$$

Use the proof, Theorem 3.8.2, and Exercise 3.8.1 to conclude that:

$$(i) \alpha_k = k^{1/\alpha}, \quad (ii) \text{ if } \alpha < 1, \text{ then } \beta_k = 0.$$

Solution. The defining scaling of an α -stable law is that adding k independent copies multiplies the scale by $k^{1/\alpha}$. Equivalently, the normalization in Theorem 3.8.2 satisfies

$$a_{kn} \sim k^{1/\alpha} a_n.$$

Therefore

$$\alpha_k = k^{1/\alpha}.$$

When $\alpha < 1$, Exercise 3.8.1 says that the centering constants may be chosen equal to 0. Hence no translation is introduced in the stable limit, and

$$\beta_k = 0.$$

Knowledge used. Stability, regular variation of normalizing constants, centering for $\alpha < 1$.

Exercise 3.8.6

Question. Let Y be a stable law with index $\alpha < 1$ and $\kappa = 1$, so that $Y \geq 0$. Let

$$\psi(\lambda) = \mathbb{E}e^{-\lambda Y}.$$

Using Exercise 3.8.5, conclude that

$$\psi(\lambda) = e^{-c\lambda^\alpha}$$

for some $c > 0$.

Solution. Exercise 3.8.5 gives, for each integer $n \geq 1$,

$$Y_1 + \dots + Y_n \stackrel{d}{=} n^{1/\alpha} Y.$$

Taking Laplace transforms,

$$\psi(\lambda)^n = \psi(n^{1/\alpha} \lambda).$$

Let $f(\lambda) = -\log \psi(\lambda)$. Then

$$f(n^{1/\alpha} \lambda) = n f(\lambda).$$

By applying this identity to ratios of integers and using continuity of f , we get

$$f(r\lambda) = r^\alpha f(\lambda), \quad r > 0.$$

Taking $\lambda = 1$ gives

$$f(\lambda) = c\lambda^\alpha, \quad c = f(1) > 0.$$

Hence

$$\psi(\lambda) = e^{-c\lambda^\alpha}.$$

Knowledge used. Laplace transforms, positive stable laws, stability scaling, continuity functional equation.

Exercise 3.8.7

Question. Show that:

(i) if X is symmetric stable with index α and $Y \geq 0$ is independent stable with index $\beta < 1$,

then $XY^{1/\alpha}$ is symmetric stable with index $\alpha\beta$.

Also let W_1, W_2 be independent standard normal random variables. Check that $1/W_2^2$ has density

$$(2\pi y^3)^{-1/2} e^{-1/(2y)}, \quad y > 0,$$

and use this to conclude that W_1/W_2 is Cauchy.

Solution. If X is symmetric α -stable, then for some $c > 0$,

$$\mathbb{E} e^{itX} = e^{-c|t|^\alpha}.$$

Conditioning on Y ,

$$\mathbb{E} e^{itXY^{1/\alpha}} = \mathbb{E} \left[e^{-c|t|^\alpha Y} \right].$$

Since Y is positive β -stable, Exercise 3.8.6 gives

$$\mathbb{E} e^{-sY} = e^{-c's^\beta}.$$

Therefore

$$\mathbb{E} e^{itXY^{1/\alpha}} = e^{-c'(c|t|^\alpha)^\beta} = e^{-C|t|^{\alpha\beta}},$$

which is the characteristic function of a symmetric stable law with index $\alpha\beta$.

For $Y = 1/W_2^2$, the transformation $w = \pm y^{-1/2}$ gives

$$f_Y(y) = 2 \frac{1}{\sqrt{2\pi}} e^{-1/(2y)} \frac{1}{2} y^{-3/2} = (2\pi y^3)^{-1/2} e^{-1/(2y)}, \quad y > 0.$$

This is the positive stable density with index $1/2$. Since W_1 is symmetric stable with index 2, part (i) implies that

$$W_1 Y^{1/2} = \frac{W_1}{|W_2|}$$

is symmetric stable with index 1, hence Cauchy. Multiplying by the independent random sign of W_2 does not change the symmetric distribution, so

$$\frac{W_1}{W_2}$$

is Cauchy.

Knowledge used. Characteristic functions, subordination of stable laws, positive one-half stable density, normal ratio representation of the Cauchy law.

Section 3.9: Infinitely Divisible Distributions

Exercise 3.9.1

Question. Show that the gamma distribution is infinitely divisible.

Solution. Let X have the gamma distribution with shape $a > 0$ and rate $\lambda > 0$. Its characteristic function is

$$\phi_X(t) = \left(\frac{\lambda}{\lambda - it} \right)^a.$$

For each integer $n \geq 1$, let $X_{n,1}, \dots, X_{n,n}$ be independent gamma random variables with shape a/n and rate λ . Then

$$\prod_{j=1}^n \phi_{X_{n,j}}(t) = \left(\frac{\lambda}{\lambda - it} \right)^a.$$

Thus X has the same distribution as $X_{n,1} + \dots + X_{n,n}$ for every n , so the gamma distribution is infinitely divisible.

Knowledge used. Gamma characteristic function, convolution of gamma laws, infinite divisibility.

Exercise 3.9.2

Question. Show that the distribution of a bounded random variable Z is infinitely divisible if and only if Z is constant. Hint: show that $\text{var}(Z) = 0$.

Solution. A constant random variable is trivially infinitely divisible. Conversely, suppose Z is bounded and infinitely divisible. If the support of Z has diameter $D < \infty$, then for each n there are independent identically distributed random variables $Y_{n,1}, \dots, Y_{n,n}$ such that

$$Z \stackrel{d}{=} Y_{n,1} + \dots + Y_{n,n}.$$

The support of an n -fold convolution has diameter n times the diameter of the support of one summand. Therefore the support diameter of $Y_{n,1}$ is at most D/n . Hence

$$\text{var}(Y_{n,1}) \leq \frac{D^2}{4n^2}.$$

Since the summands are independent,

$$\text{var}(Z) = n \text{var}(Y_{n,1}) \leq \frac{D^2}{4n}.$$

Letting $n \rightarrow \infty$ gives $\text{var}(Z) = 0$, so Z is constant almost surely.

Knowledge used. Infinite divisibility, support of convolutions, variance additivity, bounded support variance bound.

Exercise 3.9.3

Question. Show that if μ is infinitely divisible, then its characteristic function ϕ never vanishes.

Solution. Let

$$\psi(t) = |\phi(t)|^2.$$

This is the characteristic function of $X - X'$, where X and X' are independent with law μ . It is also infinitely divisible.

Assume for contradiction that $\phi(t_0) = 0$. Then $\psi(t_0) = 0$. Since ψ is infinitely divisible, for every n there is a characteristic function ψ_n such that

$$\psi(t) = \psi_n(t)^n.$$

Choose a small interval around 0 on which $\psi(t) > 0$. On this interval,

$$\psi_n(t) = \psi(t)^{1/n} \rightarrow 1.$$

By the continuity theorem, the laws with characteristic functions ψ_n converge weakly to the point mass at 0. Hence

$$\psi_n(t_0) \rightarrow 1.$$

But $\psi(t_0) = 0$ implies $\psi_n(t_0) = 0$ for every n , a contradiction. Therefore ϕ has no zeros.

Knowledge used. Characteristic functions, symmetrization, infinite divisibility, continuity theorem.

Exercise 3.9.4

Question. Find the Levy measure of the limit distribution in Exercise 3.4.13(iii).

Solution. The limit in Exercise 3.4.13(iii) has characteristic function

$$\phi(t) = \exp \left\{ \int_0^1 \frac{\cos(tx) - 1}{x} dx \right\}.$$

This is a symmetric infinitely divisible law. In the Levy-Khinchin formula the imaginary compensation terms cancel for symmetric measures. Thus we want

$$\int_{\mathbb{R}} (\cos(tx) - 1) \nu(dx) = \int_0^1 \frac{\cos(tx) - 1}{x} dx.$$

The required Levy measure is therefore

$$\nu(dx) = \frac{1}{2|x|} \mathbf{1}_{\{0 < |x| \leq 1\}} dx.$$

Indeed,

$$\int_{-1}^1 (\cos(tx) - 1) \frac{1}{2|x|} dx = \int_0^1 \frac{\cos(tx) - 1}{x} dx.$$

Knowledge used. Levy-Khinchin formula, symmetric Levy measures, characteristic exponent matching.

Section 3.10: Multidimensional Distributions

Exercise 3.10.1

Question. Show how to recover the one-dimensional marginal distribution functions from a d -dimensional distribution function F .

Solution. Let $X = (X_1, \dots, X_d)$ have distribution function F . The marginal distribution function of X_i is

$$F_i(x) = \mathbb{P}(X_i \leq x).$$

This is obtained from F by sending all other coordinates to infinity:

$$F_i(x) = \lim_{x_j \rightarrow \infty, j \neq i} F(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_d).$$

Indeed, the events

$$\{X_i \leq x, X_j \leq x_j \text{ for } j \neq i\}$$

increase to $\{X_i \leq x\}$ as the other coordinates tend to infinity.

Knowledge used. Marginal distributions, continuity from below, multidimensional distribution functions.

Exercise 3.10.2

Question. Let $F_1, \dots, F_d$ be distribution functions. Show that for $\alpha \in [-1, 1]$,

$$F(x_1, \dots, x_d) = \left\{ 1 + \alpha \prod_{i=1}^d (1 - F_i(x_i)) \right\} \prod_{j=1}^d F_j(x_j)$$

is a distribution function with marginals $F_1, \dots, F_d$.

Solution. It is enough to verify the corresponding copula on $[0, 1]^d$:

$$C(u_1, \dots, u_d) = \prod_{j=1}^d u_j + \alpha \prod_{j=1}^d u_j (1 - u_j).$$

If the u_j are continuity points, its mixed derivative is

$$\frac{\partial^d C}{\partial u_1 \cdots \partial u_d} = 1 + \alpha \prod_{j=1}^d (1 - 2u_j).$$

Since $\prod_j (1 - 2u_j) \in [-1, 1]$ and $\alpha \in [-1, 1]$, this derivative is nonnegative. Hence C is d -increasing. The formula is also right-continuous and has the correct boundary behavior.

For the marginals, if all variables except u_i are set equal to 1, then the second product vanishes and

$$C(1, \dots, 1, u_i, 1, \dots, 1) = u_i.$$

Therefore

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$$

is a joint distribution function with marginal distribution functions $F_1, \dots, F_d$.

Knowledge used. Copulas, Frechet construction, mixed derivatives, marginal recovery.

Exercise 3.10.3

Question. If a random vector has a continuous density f , show that

$$\frac{\partial^d}{\partial x_1 \cdots \partial x_d} F(x_1, \dots, x_d) = f(x_1, \dots, x_d).$$

Solution. If X has continuous density f , then

$$F(x_1, \dots, x_d) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_d} f(y_1, \dots, y_d) dy_d \cdots dy_1.$$

Applying the fundamental theorem of calculus successively in $x_d, x_{d-1}, \dots, x_1$ gives

$$\frac{\partial^d F}{\partial x_1 \cdots \partial x_d} = f(x_1, \dots, x_d),$$

at every point of continuity of f , hence everywhere under the stated continuity assumption.

Knowledge used. Joint density, distribution function, iterated integrals, fundamental theorem of calculus.

Exercise 3.10.4

Question. Show that if $X_n \Rightarrow X$ as random vectors in $\mathbb{R}^d$, then each coordinate $X_{n,i} \Rightarrow X_i$.

Solution. The coordinate projection

$$\pi_i(x_1, \dots, x_d) = x_i$$

is continuous. By the continuous mapping theorem,

$$\pi_i(X_n) \Rightarrow \pi_i(X).$$

That is,

$$X_{n,i} \Rightarrow X_i.$$

Knowledge used. Weak convergence in $\mathbb{R}^d$, coordinate projections, continuous mapping theorem.

Exercise 3.10.5

Question. If ϕ is a characteristic function on $\mathbb{R}$, describe the distribution with characteristic function

$$\psi(t_1, \dots, t_d) = \phi(t_1 + \dots + t_d).$$

Solution. Let X have characteristic function ϕ , and define the random vector

$$Y = (X, \dots, X) \in \mathbb{R}^d.$$

Then

$$\mathbb{E}e^{i(t_1 Y_1 + \dots + t_d Y_d)} = \mathbb{E}e^{i(t_1 + \dots + t_d)X} = \phi(t_1 + \dots + t_d).$$

Thus ψ is the characteristic function of the distribution concentrated on the diagonal

$$\{(x, \dots, x) : x \in \mathbb{R}\},$$

with one-dimensional law determined by ϕ .

Knowledge used. Multivariate characteristic functions, diagonal embeddings, uniqueness of characteristic functions.

Exercise 3.10.6

Question. Show that random variables $X_1, \dots, X_d$ are independent if and only if their joint characteristic function factors as

$$\phi(t_1, \dots, t_d) = \prod_{j=1}^d \phi_j(t_j),$$

where ϕ_j is the characteristic function of X_j .

Solution. If $X_1, \dots, X_d$ are independent, then

$$\phi(t_1, \dots, t_d) = \mathbb{E} \prod_{j=1}^d e^{it_j X_j} = \prod_{j=1}^d \mathbb{E} e^{it_j X_j} = \prod_{j=1}^d \phi_j(t_j).$$

Conversely, suppose the factorization holds. Let $Y_1, \dots, Y_d$ be independent random variables with $Y_j \stackrel{d}{=} X_j$. The joint characteristic function of $(Y_1, \dots, Y_d)$ is exactly $\prod_j \phi_j(t_j)$, which equals the joint characteristic function of $(X_1, \dots, X_d)$. By uniqueness of characteristic functions,

$$(X_1, \dots, X_d) \stackrel{d}{=} (Y_1, \dots, Y_d),$$

so $X_1, \dots, X_d$ are independent.

Knowledge used. Characteristic functions, independence, product measures, uniqueness theorem.

Exercise 3.10.7

Question. Show that the coordinates of a multivariate normal random vector are independent if and only if all off-diagonal covariances are zero.

Solution. Let X be multivariate normal with mean vector θ and covariance matrix Γ . Its characteristic function is

$$\phi(t) = \exp \left\{ it\theta^T - \frac{1}{2}t\Gamma t^T \right\}.$$

If the coordinates are independent, then their covariances are zero.

Conversely, if all off-diagonal covariances are zero, then Γ is diagonal. Therefore

$$t\Gamma t^T = \sum_{j=1}^d \Gamma_{jj} t_j^2,$$

and

$$\phi(t) = \prod_{j=1}^d \exp \left\{ i\theta_j t_j - \frac{1}{2}\Gamma_{jj} t_j^2 \right\}.$$

This is the product of the marginal characteristic functions. By Exercise 3.10.6, the coordinates are independent.

Knowledge used. Multivariate normal characteristic function, covariance matrix, independence criterion by characteristic functions.

Exercise 3.10.8

Question. Show that X is multivariate normal with mean vector θ and covariance matrix Γ if and only if every linear combination cX^T is normal with mean $c\theta^T$ and variance $c\Gamma c^T$.

Solution. If X is multivariate normal with mean θ and covariance Γ , then for every row vector c ,

$$cX^T$$

is normal with mean $c\theta^T$ and variance $c\Gamma c^T$ by the definition and characteristic function of the multivariate normal.

Conversely, suppose every linear combination cX^T is normal with mean $c\theta^T$ and variance $c\Gamma c^T$. Then for every $t \in \mathbb{R}^d$,

$$\mathbb{E} e^{itX^T} = \mathbb{E} e^{i(tX^T)} = \exp \left\{ it\theta^T - \frac{1}{2}t\Gamma t^T \right\}.$$

This is the characteristic function of the multivariate normal $N(\theta, \Gamma)$. By uniqueness of characteristic functions, X has that multivariate normal distribution.

Knowledge used. Cramer-Wold viewpoint, linear combinations, multivariate normal characteristic function, uniqueness theorem.